

Minkowski Bounds for Pasten’s Arithmetic Derivative Lattices in the Squarefree Subfamily

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Abstract

For each coprime triple (a, b, c) with $a + b = c$, Pasten (2021) constructs a *universal derivative lattice* $F(a, b)$ of rank $\omega(abc) - 1$ from the linear constraint imposed by the weighted arithmetic derivative. We prove that for *squarefree* coprime triples, the lattice determinant satisfies $\det(F(a, b)) < \text{rad}(abc)$. Combined with the Minkowski–Vaaler theorem, this yields a nonzero lattice vector with ℓ^∞ -norm $\|\psi\|_\infty \leq \text{rad}(abc)^{1/(\omega-1)}$.

We give a complete classification of partition types by whether the minimum *non-degenerate* norm ratio $\rho = \|\psi_{\text{nd}}\|_\infty / R^{1/(\omega-1)}$ is universally bounded: nine types are bounded (with exact analytical sharp bounds including $2^{-1/(\omega-1)}$ for balanced types at $\omega = 3, 4, 5$, $\sup \rho = 1$ for two $\omega = 5$ types with $a = 1$, $\sup \rho = 2^{-1/4}$ for type $(2, 1, 2)$, and $\sup \rho = 6^{-1/4}$ for type $(2, 2, 1)$). All $\omega \geq 6$ types are unbounded. All nine bounded $\omega = 5$ types have asymptotic suprema (never achieved). All sharp bounds have been verified over 44 000 squarefree triples and the key inequalities formalized in Lean 4 (zero `sorry`).

We prove a clean quality boundary: for squarefree triples with $a \leq b$, quality $> 1/2$ if and only if $a = 1$. Combined with the classification, this shows that the triples hardest for abc (highest quality) are structurally opposite to the triples closest to the sharp ρ bound. The argument is self-contained and uses no abc-equivalent hypothesis.

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Honest scope. The result proved here is a structural property of Pasten’s lattice. It does *not* prove the abc conjecture, nor does it provide progress toward abc for the hard cases (high-quality triples, which have small ω). The abc conjecture remains [OUT] in the sense of the accompanying verification kernel.

1 Introduction

1.1 Background

The abc conjecture of Masser–Oesterlé asserts that for every $\varepsilon > 0$ there exists $K_\varepsilon > 0$ such that for all coprime positive integers a, b, c with $a + b = c$:

$$c \leq K_\varepsilon \cdot \text{rad}(abc)^{1+\varepsilon},$$

where $\text{rad}(n) = \prod_{p|n} p$ is the radical. This remains an open problem.

Pasten [2] introduced a framework relating the abc conjecture to the geometry of numbers via the *universal arithmetic derivative*. For each prime p and weight $\psi(p) \in \mathbb{Z}$, the ψ -arithmetic derivative is

$$d^\psi(n) = n \sum_{p|n} \frac{v_p(n)}{p} \psi(p),$$

where $v_p(n)$ is the p -adic valuation. The key object is the *lattice of additive weights*:

$$F(a, b) = \{ \psi : \mathbb{P}(abc) \rightarrow \mathbb{Z} \mid d^\psi(a) + d^\psi(b) = d^\psi(c) \}.$$

Since $\gcd(a, b) = 1$ and $a + b = c$, the prime supports P_a, P_b, P_c of a, b, c are pairwise disjoint (for squarefree triples), and the additivity constraint reduces to a single integer linear equation. Pasten showed that $F(a, b)$ has rank $\omega(abc) - 1$, where $\omega(n)$ denotes the number of distinct prime divisors of n .

Pasten's *Small Derivatives Conjecture* (SDC) asserts that the minimum ℓ^∞ -norm of a non-degenerate vector in $F(a, b)$ is at most c^η for some $\eta < 1$. Pasten proved that SDC is equivalent to the abc conjecture. Siegel's lemma gives only $\eta = 1$ (the trivial bound); beating this would be a breakthrough.

Our contribution is more modest: we establish the exact determinant of $F(a, b)$ for squarefree triples, and derive the corresponding Minkowski bound $\|\psi\|_\infty \leq \text{rad}(abc)^{1/(\omega-1)}$. This is the $\eta = 1/(\omega-1)$ *bound relative to* $\text{rad}(abc)$, which is non-trivial when $\omega \geq 3$. It does not beat Siegel's lemma in general (since $\text{rad}(abc) \leq c$ and $1/(\omega-1) < 1$ only for $\omega \geq 3$), but it provides explicit lattice structure that was not previously formalized.

1.2 Main results

Let (a, b, c) be a coprime squarefree triple with $a + b = c$ and $\omega = \omega(abc) \geq 2$. Set $P = P_a \cup P_b \cup P_c$ and $R = \text{rad}(abc) = \prod_{p \in P} p$.

Theorem 1 (Determinant bound). $\det(F(a, b)) = R \cdot \sqrt{\sum_{p \in P} p^{-2}} < R$.

Theorem 2 (GCD lemma). *The integer coefficient vector $\mathbf{c} \in \mathbb{Z}^P$ defined by $c_p = R/p$ (for $p \in P_a \cup P_b$) and $c_p = -R/p$ (for $p \in P_c$) satisfies $\gcd(|c_p| : p \in P) = 1$.*

Corollary 3 (Minkowski–Vaaler bound [1, 3]). *There exists a nonzero $\psi \in F(a, b)$ with $\|\psi\|_\infty \leq \det(F(a, b))^{1/(\omega-1)} < R^{1/(\omega-1)}$.*

For non-squarefree triples we prove the following extension (Theorem 15 below): the minimum non-degenerate norm satisfies $\text{nd}(a, b) \leq v_{\max} \cdot R^{1/(\omega^*-1)}$, where $v_{\max} = \max_p v_p(abc)$. We also establish a universal lower bound (Theorem 19): $\text{nd}(a, b) \geq p_2$, the second-smallest prime in $\text{rad}(abc)$, for every coprime triple regardless of squarefreeness; and an exact formula for $\omega^* = 2$ (Theorem 20).

Corollary 3 does *not* imply abc for the following reasons:

1. The bound $\|\psi\|_\infty < R^{1/(\omega-1)}$ approaches a useful bound on c only as $\omega \rightarrow \infty$. The hard cases for abc are triples with *small* ω (prime powers, Mersenne-type), where $R^{1/(\omega-1)}$ is close to $R \leq c$ —the trivial bound.
2. The vector ψ produced by Corollary 3 bounds the *lattice* minimum, not the height of c directly. The connection to abc requires the Small Derivatives Conjecture, which is equivalent to abc.
3. For the degenerate $\omega = 2$ family (e.g. $c = 2^k$, $a = 1$), the bound gives $\|\psi\|_\infty \leq R^1 = R$, which is the trivial bound. Pasten explicitly excludes this family from SDC for this reason.

2 Definitions and setup

2.1 The Pasten lattice for squarefree triples

Let (a, b, c) be a coprime squarefree triple, $a + b = c$, $a, b, c > 0$. Write $v_p(n) = 1$ for all $p \mid n$ (squarefree hypothesis).

Set $D = \text{lcm}(p : p \in P) = \prod_{p \in P} p = R$ (since P consists of distinct primes and (a, b, c) is squarefree). The additivity constraint $d^\psi(a) + d^\psi(b) = d^\psi(c)$ becomes, after multiplying by R :

$$\sum_{p \in P} c_p \cdot \psi(p) = 0,$$

where

$$c_p = \begin{cases} +R/p & p \in P_a \cup P_b, \\ -R/p & p \in P_c. \end{cases}$$

Note $c_p \in \mathbb{Z}$ since $p \mid R$. The lattice is

$$F(a, b) = L = \{\psi \in \mathbb{Z}^P \mid \mathbf{c} \cdot \psi = 0\},$$

a rank- $(\omega - 1)$ sublattice of $\mathbb{Z}^P$.

2.2 Lattice determinant

For an integer lattice defined by a single primitive constraint $\mathbf{c} \in \mathbb{Z}^\omega$ with $\text{gcd}(c_p) = 1$, the determinant equals

$$\det(L) = \|\mathbf{c}\|_2 = \left(\sum_{p \in P} c_p^2 \right)^{1/2}.$$

If $\text{gcd}(c_p) = g > 1$, the primitive form is $\mathbf{c}/g$ and $\det(L) = \|\mathbf{c}\|_2/g$.

3 Proofs

3.1 Proof of Theorem 2

Proof. Each $|c_p| = R/p = \prod_{q \in P, q \neq p} q$. Suppose a prime ℓ divides $\text{gcd}(|c_p| : p \in P)$. Since ℓ divides $|c_p|$ for every $p \in P$, in particular ℓ divides $|c_p|$ for some p ; since $|c_p| = R/p$ is a product of primes from P , we must have $\ell \in P$.

Take $p = \ell$: then $|c_\ell| = R/\ell = \prod_{q \in P, q \neq \ell} q$. This product does *not* contain ℓ as a factor, so $\ell \nmid R/\ell$. But ℓ was assumed to divide every $|c_p|$, including $|c_\ell|$. Contradiction. Hence $\text{gcd}(|c_p|) = 1$. $\square$

Remark 1. *Edge case* $|P| = 2$, $P = \{p, q\}$: $|c_p| = q$, $|c_q| = p$, $\text{gcd}(p, q) = 1$. *The case* $|P| = 1$ *is excluded by* $\omega \geq 2$.

3.2 Proof of Theorem 1

Proof. By Theorem 2, $\text{gcd}(|c_p| : p \in P) = 1$, so $\det(L) = \|\mathbf{c}\|_2$. Since $c_p = \pm R/p$ we have $c_p^2 = (R/p)^2 = R^2/p^2$, thus

$$\|\mathbf{c}\|_2^2 = \sum_{p \in P} \frac{R^2}{p^2} = R^2 \sum_{p \in P} \frac{1}{p^2}.$$

So $\det(L) = R \sqrt{\sum_{p \in P} p^{-2}}$.

It remains to show $\sum_{p \in P} p^{-2} < 1$. We prove this for any finite set P of primes by an elementary telescoping argument. Set $M = \max P$. For each $n \geq 2$:

$$\frac{1}{n^2} \leq \frac{1}{n-1} - \frac{1}{n},$$

which follows from the identity $\frac{1}{n-1} - \frac{1}{n} - \frac{1}{n^2} = \frac{1}{n^2(n-1)} > 0$. Therefore

$$\sum_{p \in P} \frac{1}{p^2} \leq \sum_{p \in P} \left(\frac{1}{p-1} - \frac{1}{p} \right) \leq \sum_{n=2}^M \left(\frac{1}{n-1} - \frac{1}{n} \right) = 1 - \frac{1}{M} < 1.$$

(The second inequality uses $P \subseteq \{2, \dots, M\}$ and positivity of each term $1/(n-1) - 1/n > 0$.) Hence $\det(L) < R$. $\square$

Remark 2. The true value is $\sum_{p \text{ prime}} p^{-2} = P(2) \approx 0.4522$ (prime zeta function at $s = 2$). An elementary bound gives $\sum_{p \text{ prime}} p^{-2} \leq 11/18 \approx 0.611 < 1$ via $1/4 + 1/9 + \int_4^\infty x^{-2} dx = 1/4 + 1/9 + 1/4 = 11/18$.

3.3 Proof of Corollary 3

Proof. By Theorem 1, $\det(L) < R$. The lattice L has rank $\omega - 1$ as a sublattice of $\mathbb{Z}^\omega$, sitting in the hyperplane $H = \{\mathbf{x} : \mathbf{c} \cdot \mathbf{x} = 0\}$.

Minkowski's convex-body theorem [1] gives a nonzero $\psi \in L$ with ℓ^∞ -norm bounded in some orthonormal basis of H . To recover the bound in the *ambient* $\mathbb{Z}^\omega$ coordinates, we additionally invoke Vaaler's theorem [3]: every central hyperplane section of $[-1, 1]^\omega$ has $(\omega - 1)$ -dimensional volume $\geq 2^{\omega-1}$. Together these give a nonzero $\psi \in L$ with

$$\|\psi\|_\infty \leq \det(L)^{1/(\omega-1)} < R^{1/(\omega-1)}.$$

$\square$

4 Numerical evidence

Table 1 gives comprehensive statistics from script `t30_rho_distribution.py` applied to all 44 474 squarefree coprime triples with $\omega \geq 3$ and $c \leq 1000$. Table 2 breaks out the seven universally-bounded partition types.

The unbounded types drive the high mean ρ for $\omega \leq 5$. For the bounded partition types, the distribution concentrates well below the sharp bounds: the median ρ for combined $\omega = 4$ bounded types is 0.34, and fewer than 1% of triples come within 0.02 of the sharp bound $2^{-1/3}$.

5 Discussion

5.1 The degenerate $\omega = 2$ family

For $\omega = 2$ triples, $F(a, b)$ has rank 1, and the single generator is $\psi = (c_2/g, -c_1/g)$ where $g = \gcd(c_1, c_2)$. In the squarefree case $c = \{p, q\}$ and the minimum norm is $\max(q, p)/1 =$

ω	count	mean ρ	max ρ	% > 1	note
3	79	0.864	1.080	30.4%	includes unbounded types (0, 2, 1), etc.
4	3 223	0.945	9.919	27.7%	(2, 1, 1), (0, 1, 3), (0, 3, 1) unbounded
5	14 009	0.961	31.105	24.1%	(1, 1, 3), (1, 3, 1), (3, 1, 1) unbounded
6	18 739	0.465	17.889	5.3%	all types unbounded
7	7 766	0.248	1.631	1.0%	all types unbounded

Table 1: Empirical summary of $\rho = \|\psi_{\text{nd}}\|_{\infty}/R^{1/(\omega-1)}$ organized by $\omega = \omega(abc)$ over all squarefree coprime triples with $c \leq 1000$. High mean ρ for $\omega = 3, 4, 5$ is driven by unbounded partition types.

Type	n	mean ρ	max ρ	sup ρ	% > 0.5	status
(1, 1, 1), $\omega = 3$	35	0.693	0.706	$2^{-1/2} \approx 0.707$	100%	BOUNDED
(0, 2, 2), $\omega = 4$	69	0.147	0.505	$3 \cdot 2 \cdot 10^{-1/3} \approx 0.505$	1.4%	BOUNDED
(1, 1, 2), $\omega = 4$	1 029	0.356	0.781	$2^{-1/3} \approx 0.794$	30.0%	BOUNDED
(1, 2, 1), $\omega = 4$	1 036	0.372	0.791	$2^{-1/3} \approx 0.794$	31.9%	BOUNDED
(1, 2, 2), $\omega = 5$	3 450	0.104	0.817	$2^{-1/4} \approx 0.8408$ (not achieved)	0.0%	BOUNDED
(2, 1, 2), $\omega = 5$	3 115	0.093	0.634	$2^{-1/4} \approx 0.841$ (not achieved)	0.3%	BOUNDED
(2, 2, 1), $\omega = 5$	3 006	0.089	0.611	$6^{-1/4} \approx 0.6389$ (not achieved)	0.4%	BOUNDED
(0, 2, 3), $\omega = 5$	—	—	—	sup = 1 (not achieved)	—	BOUNDED
(0, 3, 2), $\omega = 5$	—	—	—	sup = 1 (not achieved)	—	BOUNDED

Table 2: Distribution of ρ for the nine universally-bounded partition types, $c \leq 1000$ (for $\omega \leq 4$ and $\omega = 5$ multi-group types). Types (0, 2, 3) and (0, 3, 2) have $a = 1$ and two non-empty prime groups; their $\text{sup } \rho = 1$ is proved analytically (Theorem 10) but never achieved. Type (2, 2, 1) has $\text{sup } \rho = 6^{-1/4} \approx 0.6389$ (Theorem 9), approached via $a = 6$, near-twin-prime $b = q_1 q_2$, $c = 6 + b$ prime; never achieved. All other maxima stay below the proved sharp bounds. P99 and P100 for combined $\omega = 4$ bounded types: 0.775 and 0.791, both below $2^{-1/3} = 0.794$.

$\max(p, q)$. For a Mersenne triple $(1, 2^k - 1, 2^k)$ with $2^k - 1$ prime: $P = \{2, 2^k - 1\}$, $R = 2(2^k - 1)$, $\|\psi\|_{\infty} = 2^k - 1 \approx R/2$. So $\|\psi\|_{\infty}/R \rightarrow 1/2$, not smaller. The bound is essentially tight.

5.2 Why this does not prove abc

Let (a, b, c) be a high-quality abc triple (quality = $\log c / \log R > 1$). High-quality triples tend to have small ω (often $\omega = 3$ or 4). Corollary 3 gives $\|\psi\|_{\infty} \leq R^{1/(\omega-1)}$. For $\omega = 3$: $\|\psi\|_{\infty} \leq R^{1/2}$. Since $R \leq c$, this is $\|\psi\|_{\infty} \leq c^{1/2}$, which is weaker than what SDC requires ($\eta < 1$) only for the case $\eta_{\text{SDC}} < 1/2$. The abc conjecture is equivalent to SDC holding for all $\varepsilon > 0$, i.e. $\|\psi\|_{\infty} = o(c^{\varepsilon})$ for all $\varepsilon > 0$ — an exponential improvement over what we prove.

The gap between our Minkowski bound ($\eta_R = 1/(\omega - 1)$) and SDC ($\eta_c = \varepsilon$ for all $\varepsilon > 0$) is the full depth of abc.

5.3 Quality boundary for squarefree triples

For squarefree coprime (a, b, c) with $a + b = c$ and $R = \text{rad}(abc) = abc$, the quality satisfies $\text{quality} = \log c / \log(abc)$. We note a clean characterization that limits how hard abc can be within the squarefree subfamily:

Proposition 4 (Quality boundary). *For squarefree coprime (a, b, c) with $a + b = c$ and $a \leq b$:*

$$\text{quality} > \frac{1}{2} \iff a = 1.$$

Moreover, $\text{quality} < 1$ always, and $\text{quality} = \frac{1}{2}$ is never achieved. For $a = 1$, $\text{quality} \rightarrow \frac{1}{2}$ from above as $b \rightarrow \infty$. For $a = 2$, $\text{quality} \rightarrow \frac{1}{2}$ from below.

Proof. $\text{quality} > \frac{1}{2}$ iff $\log c > \log a + \log b$, i.e. $c > ab$, i.e. $a + b > ab$, i.e. $(a - 1)(b - 1) < 1$. For positive integers, this means $(a - 1)(b - 1) = 0$, i.e. $a = 1$ or $b = 1$. Since $a \leq b$ and $a \geq 1$: $b = 1 \Rightarrow a = 1$. So $\text{quality} > 1/2$ iff $a = 1$. $\text{Quality} < 1$ since $R = abc > c$. $\text{Quality} = 1/2$ would require $c = ab$, impossible since $c = a + b < ab$ for $a, b \geq 2$ and $c = a + b > ab = b$ for $a = 1$. $\square$

The Lean 4 formalization of the combinatorial core ($ab < a + b \leftrightarrow a = 1$ for $1 \leq a \leq b$) is given by `pasten.F21A.ab_lt_sum_iff_a_one` (proved by `omega/nlinarith`, zero sorry).

5.4 Quality- ρ trade-off

Proposition 4 implies that squarefree triples with $\text{quality} > 1/2$ are exactly the *trivial* triples ($a = 1$). For the harder non-trivial cases ($a \geq 2$), $\text{quality} \leq 1/2$ with equality only in the limit.

Within bounded partition types, quality and ρ point in *opposite* directions:

- **High- ρ families** (balanced types approaching the sharp bound): e.g. type $(1, 1, 2)$ with $a = p \approx q$ both large odd primes. Here $\rho^3 \rightarrow 1/2$ (approaching $2^{-1/3}$) while $\text{quality} \rightarrow \log(2p) / \log(2p^3) = 1/3$.
- **High-quality families** ($a = 2$ fixed, $b \rightarrow \infty$): Here $\text{quality} \rightarrow 1/2$ (approaching the squarefree non-trivial supremum) while $\rho = 2 / (2 \cdot b \cdot (2 + b))^{1/3} \rightarrow 0$.

Empirically (Table 2, $c \leq 2000$), Pearson $r(\text{quality}, \rho) \approx -0.76$ for types $(1, 1, 2)$ and $(1, 2, 1)$.

Conclusion. The triples hardest for abc (highest quality, approaching $\text{quality} = 1/2$ from below) are not the triples closest to the sharp ρ bound. The gap between “hard for abc ” and “hard for SDC” is a provable structural feature: within the squarefree subfamily, maximizing quality and maximizing ρ are *geometrically opposite* in the parameter space of partition types.

F31: joint quality- ρ bound

The opposition between quality and ρ is not merely empirical: for every *bounded- ρ* partition type (those with $\sup_{\text{triples}} \rho < \infty$) the sum $\text{quality} + \rho^{\omega-1}$ is provably bounded away from 1.

Theorem 5 (F31: Quality- ρ Joint Bound). *For every squarefree coprime triple (a, b, c) with $a + b = c$, $a \geq 2$, belonging to a bounded- ρ partition type:*

$$\text{quality}(a, b, c) + \rho(a, b, c)^{\omega-1} < 1.$$

The supremum equals 1 and is approached only by type $(1, 1, 1)$ triples as $p \rightarrow \infty$.

Proof. By Proposition 4 (quality boundary), $\text{quality} < 1/2$ for all non-trivial squarefree triples ($a \geq 2$). By Theorem 35 (sharp ρ bounds), every bounded- ρ type has $\sup \rho \leq 2^{-1/(\omega-1)}$, so $\rho^{\omega-1} \leq (2^{-1/(\omega-1)})^{\omega-1} = 1/2$ (strict: the supremum is never achieved). Hence $\text{quality} + \rho^{\omega-1} < 1/2 + 1/2 = 1$. $\square$

Remark 3 (Tightness for type $(1, 1, 1)$). *For the twin-prime family $(2, q, q + 2)$: $\text{quality} \rightarrow 1/2$ and $\rho^2 = q/(2(q + 2)) \rightarrow 1/2$ as $q \rightarrow \infty$, so the bound is tight. The algebraic core — $(q + 2)^q < (2q)^{q+4}$ for all $q \geq 1$ (proved for $q = 1$ directly; for $q \geq 2$ by $q + 2 \leq 2q$ and $q < q + 4$) — is formalized as `pasten_F31a` in the Lean kernel.*

Remark 4 (Unbounded types). *The bound fails for unbounded- ρ types (e.g. type $(2, 1, 1)$: $\rho^3 \approx q/6 \rightarrow \infty$). For bounded types the gap to 1 is large away from type $(1, 1, 1)$: $\omega = 4$ bounded types maximum ≈ 0.523 ; $\omega = 5$ bounded types ≈ 0.569 ; $\omega = 6$ bounded types ≈ 0.889 (all verified numerically, T51).*

5.5 Non-degeneracy: complete classification for $\omega \leq 5$

Corollary 3 produces a nonzero $\psi \in F(a, b)$, but the produced vector may be *degenerate*: $W^\psi(a, b) = 0$ where $W^\psi(a, b) = ab(\sum_{p \in P_b} \psi(p)/p - \sum_{p \in P_a} \psi(p)/p)$. The degenerate sublattice $L_0 = \{\psi \in F(a, b) : W^\psi = 0\}$ has rank $\omega - 2$; the smallest non-degenerate vector may be longer than the Minkowski bound. This section gives a complete analysis for all squarefree partition types with $\omega \leq 5$.

Universal Divisibility Lemma and coordinate change

Theorem 6 (F8: Universal Divisibility). *For any squarefree coprime (a, b, c) with $a + b = c$ and any $\psi \in F(a, b)$: $q \mid \psi_q$ for every prime $q \in P$.*

Proof. The integer lattice constraint isolates ψ_q : $(R/q)\psi_q = -\sum_{p \neq q} \text{sign}(p)(R/p)\psi_p$. For each $p \neq q$: R/p contains q as a factor (since $q \in P$, $q \neq p$). Hence $q \mid \text{RHS}$. Since $\gcd(q, R/q) = 1$: $q \mid \psi_q$. $\square$

Define $\varphi_q = \psi_q/q \in \mathbb{Z}$ (well-defined by Theorem 6). This gives a canonical bijection

$$F(a, b) \cong \tilde{F}(a, b) = \{\varphi \in \mathbb{Z}^\omega \mid \sum_{q \in P} \text{sign}(q) \varphi_q = 0\},$$

where $\text{sign}(q) = +1$ if $q \mid ab$ and -1 if $q \mid c$. The Wronskian becomes $W = ab(S_b - S_a)$ where $S_X = \sum_{q \mid X} \varphi_q$. Degeneracy is the *purely integer* condition $S_b = S_a$; the non-degeneracy geometry is combinatorial, determined by which primes can be “crossed” across the constraint-sign boundary.

$\omega = 3$: all non-degenerate

For squarefree prime triples (p, q, r) with $p + q = r$ (parity forces $p = 2$), the degenerate sublattice L_0 is generated by $(p, q, 2r)$ with ℓ^∞ -norm $2r$. Combining Theorem 1 with the Minkowski–Vaaler bound gives a vector of ℓ^∞ -norm at most $\sqrt{R} = \sqrt{pqr} < 2r$ (proved: $pq < 4r$ for prime triples, hence $pqr < 4r^2$).

Theorem 7 (F3: $\omega=3$ non-degeneracy). *For every squarefree prime triple (p, q, r) with $p + q = r$, the shortest non-zero vector of $F(p, q)$ is non-degenerate; equivalently, $\text{nd}(p, q) < 2r$.*

Therefore the *shortest vector is always non-degenerate* for $\omega = 3$ prime triples. This is Theorem 7; four constituent lemmas are formalized in Lean 4 (zero `sorry`).

$\omega = 4$: complete partition-type classification

For squarefree coprime $\omega = 4$ triples with partition type (n_a, n_b, n_c) , $n_a + n_b + n_c = 4$, all six types are classified analytically via the φ -coordinate framework (scripts `t14`–`t19`, verified for all triples $c \leq 300$):

Type	Degen?	Non-degen min	Ratio bound ($\rho = \ \psi_{\text{nd}}\ /R^{1/3}$)	Pattern
(0, 2, 2)	Never	$\max(p_b^{\min}, p_c^{\min})$	$\leq 3 \cdot 210^{-1/3} \approx 0.505$, max at (1, 14, 15)	BOUNDED
(1, 1, 2)	Never	$\max(p_a, p_b)$	$\sup = 2^{-1/3} \approx 0.794$ (F14)	BOUNDED
(1, 2, 1)	iff $a > r$	$\max(a, q)$	$\sup = 2^{-1/3} \approx 0.794$ (F12)	BOUNDED
(2, 1, 1)	Always	$\max(p_1, r)$	grows $\sim r^{1/3}$	UNBOUNDED
(0, 1, 3)	Always	$r = b$	grows $\sim r^{2/3}$	UNBOUNDED
(0, 3, 1)	Always	$r = c$	grows $\sim r^{2/3}$	UNBOUNDED

Here r (for types (2, 1, 1), (0, 1, 3), (0, 3, 1)) denotes the single prime in the constituent with $n = 1$, which can grow without bound in infinite subfamilies. For type (1, 2, 1), the degeneracy condition $a > r$ (where r is the larger prime of b) forces $r < a \leq 2r$, and the ratio $\max(a, q)/R^{1/3}$ is bounded by $2^{-1/3}$ (proved by elementary calculus: the function $t^{2/3}/(2(t+2))^{1/3}$ is strictly increasing on $(1, 2)$ with supremum $2^{-1/3}$ at $t \rightarrow 2$).

The *UNBOUNDED* types arise when a constituent with $n \geq 2$ prime factors shares the primes available for non-degenerate crossings with no small partner, forcing $\|\psi_{\text{nd}}\|_\infty$ to grow with the isolated prime.

$\omega = 5$: all partition types

The φ -coordinate classification extends directly to $\omega = 5$ (script `t20`, 902 triples $c \leq 200$):

Type	Degen/ND	Max ρ ($= \ \psi_{\text{nd}}\ /R^{1/4}$)	Achieved at	Pattern
(1, 2, 2)	0/219	$2^{-1/4} \approx 0.8408$ (sup)	—	BOUNDED, sup= $2^{-1/4}$ (F26)
(2, 1, 2)	17/171	$2^{-1/4} \approx 0.841$ (sup)	—	BOUNDED, sup= $2^{-1/4}$ (F30)
(2, 2, 1)	15/189	$6^{-1/4} \approx 0.6389$ (sup)	—	BOUNDED, sup= $6^{-1/4}$ (F24)
(0, 2, 3)	—	< 1 (sup = 1)	—	BOUNDED, sup= 1 (F23)
(0, 3, 2)	—	< 1 (sup = 1)	—	BOUNDED, sup= 1 (F23)
(1, 1, 3)	112/118	2.798	—	UNBOUNDED (F11)
(1, 3, 1)	106/110	2.798	—	UNBOUNDED (F11)
(3, 1, 1)	0/123	6.745	—	UNBOUNDED (F11)
(0, 1, 4)	—	$\rightarrow \infty$	—	UNBOUNDED (F22)
(0, 4, 1)	—	$\rightarrow \infty$	—	UNBOUNDED (F22)

All five bounded $\omega = 5$ types have *asymptotic* suprema (never achieved at a finite triple). Type (1, 2, 2) extends the universal bound $2^{-1/(\omega-1)}$ to $\omega = 5$: for $a = 2$, $b = pq$, $c = rs$ with all four odd primes near equal and $2 + pq = rs$, $\rho^4 = p^3/(2qrs) < 1/2$ since $q, r, s > p$ (Theorem 8). As all four primes $\rightarrow n$: $\rho^4 \rightarrow 1/2$; $\sup \rho = 2^{-1/4} \approx 0.8408$. Verified: 0 violations of $\rho < 2^{-1/4}$ for $a = 2$, $b \leq 200\,000$. Types (2, 1, 2) and (1, 2, 2) share $\sup \rho = 2^{-1/4} \approx 0.8409$: by the $a \leftrightarrow b$ symmetry of ρ (which depends only on $\{\min P_a, \min P_b, \min P_c\}$ and R), any (2, 1, 2) triple (a, b, c) gives a (1, 2, 2) triple (b, a, c) with the same ρ . Hence $\sup \rho$ for (2, 1, 2) equals $\sup \rho$ for (1, 2, 2), proved $= 2^{-1/4}$ by F26. The extremal family: $b = 2$ (fixed), $a = p_1 p_2$, $c = p_1 p_2 + 2 = r_1 r_2$ near-twin semiprimes; $\text{nd} = p_1$, $\rho^4 = p_1^3/(2 p_2 r_1 r_2) < 1/2$ (since $p_2 r_1 r_2 \geq p_1 p_2^2 > p_1^3$). As $p_1 \approx p_2 \approx r_1 \approx r_2 \rightarrow n$: $\rho^4 \rightarrow 1/2$. Verified: 0 violations of $\rho < 2^{-1/4}$ for 2,475,742 triples at $c \leq 50,000$ (6 violations of the old incorrect bound $6^{-1/4}$; the old F25 claim was wrong).

Theorem 8 (F26: type (1, 2, 2) bound). *For any type (1, 2, 2) triple with $a = 2$, $b = pq$, $c = rs$ (all four p, q, r, s odd primes, $2 + pq = rs$, $p \leq q$, $r \leq s$): $\rho^4 = p^3/(2qrs) < 1/2$, so $\rho < 2^{-1/4}$. This is the $\omega = 5$ case of Theorem 36 (type (1, 2, 2) = $(1, k_1, k_2)$ with $k_1 = k_2 = 2$); $\sup \rho = 2^{-1/4}$ is never achieved.*

Theorem 9 (F24: type (2, 2, 1) bound). *For any type (2, 2, 1) triple: $\rho^4 < 1/6$, so $\sup \rho \leq 6^{-1/4} \approx 0.6389$. The supremum is never achieved; it is approached by the $a = 6$ extremal family (near-twin-prime $b = q_1 q_2$, $c = 6 + b$ prime). (Lean: `pasten_F29_221_rho4_lt_sixth`, which proves the general bound F29 covering all type (2, 2, 1) triples.)*

Theorem 10 (F23: types (0, 2, 3) and (0, 3, 2) bound). *For any type (0, 2, 3) or (0, 3, 2) triple ($a = 1$, two non-empty prime groups): with $p = \min P_b$ and $q = \min P_c$ (or vice versa), $p < q$:*

$$\rho^4 = \frac{p^3}{q(pq + 1)} < \frac{p^2}{q^2} < 1,$$

so $\rho < 1$ for every finite triple. Yet $\sup \rho = 1$, approached as $p, q \rightarrow n$ near-equal.

Type (2, 2, 1) has $\sup \rho = 6^{-1/4} \approx 0.6389$ (Theorem 9, general bound F29), approached but never achieved by the $a = 6$ extremal family (see above). Types (0, 2, 3) and (0, 3, 2) have $a = 1$ (empty P_a), giving only two non-empty prime groups; their F10 $\text{nd} = \max(\min P_b, \min P_c)$ satisfies $\rho < 1$ always (Theorem 10: $\rho^4 = p^3/(q(pq + 1)) < (p/q)^2 < 1$ for $p < q$), but $\sup \rho = 1$ approached by near-equal prime pairs. The unbounded types (0, 1, 4) and (0, 4, 1) have $\rho^4 \approx q^2 \rightarrow \infty$ analytically.

F10: optimal crossing formula (general ω)

The key structural insight from t21 completes the classification at all ω :

Theorem 11 (F10, proved 2026-08-15). *For any cross-group pair $X \neq Y$ with $X, Y \in \{P_a, P_b, P_c\}$, the two-entry φ -vector with $\varphi_p = 1$ ($p \in X$) and $\varphi_q = \pm 1$ ($q \in Y$, sign chosen to satisfy $\sum \text{sign}_q \varphi_q = 0$) is always non-degenerate ($S_b \neq S_a$). Its ψ -norm is $\max(p, q)$.*

Consequently, the minimum non-degenerate norm satisfies

$$\|\psi_{\text{nd}}\|_\infty = \min_{X \neq Y} \max(\min X, \min Y) = \text{second smallest of } \{\min P_a, \min P_b, \min P_c\},$$

where $\min \emptyset = \infty$.

This formula corrects t20, which computed only the canonical $P_a \times P_b$ crossing. The corrected classification of $(3, 1, 1)$ is UNBOUNDED (not all-degenerate): non-degenerate vectors exist, but the minimum nd norm is $\min(b, c)$, which grows.

E10/E11/ E_n : explicit minimum vectors and universal lower bound

Theorem 11 asserts an equality; we now give the explicit achieving vector and prove the matching lower bound, completing the characterisation.

Theorem 12 (E10). *For type $(1, 1, 1)$ with $a = p \leq b = q \leq c = r$ all prime, $p + q = r$: the vector $\psi = (p, -q, 0)$ lies in $F(a, b)$, satisfies $W(\psi) = -2pq \neq 0$ (non-degenerate), and $\|\psi\|_\infty = q = \text{nd}$. (Lean: `pasten_E10_vec_in_lattice`, `nondeg`, `norm`.)*

Theorem 13 (E11). *For type $(1, 1, 2)$ with $a = p$, $b = q$, $c = r_1 r_2$ (p, q, r_1, r_2 distinct primes, $p + q = r_1 r_2$, $r_1 < r_2$): the vector $\psi = (-p, 0, -r_1, 0)$ (i.e., $\varphi = (-1, 0, -1, 0)$ in the F8 φ -coordinates) lies in $F(a, b)$, has $W(\psi) = pq \neq 0$, and $\|\psi\|_\infty = \max(p, r_1) = \text{nd}$. Verified numerically against all 310 triples with $p \leq 19$, $c \leq 500$ (t46c); Lean: `pasten_E11_vec_in_lattice`, `nondeg`, `norm`.*

Theorem 14 (E_n : universal lower bound). *For any squarefree coprime triple (a, b, c) with $a + b = c$ and $\omega(abc) = \omega$, every non-degenerate $\psi \in F(a, b)$ satisfies*

$$\|\psi\|_\infty \geq \text{nd} = \text{second smallest of } \{\min P_a, \min P_b, \min P_c\}.$$

Proof. By the Universal Divisibility Lemma (Theorem 2), $\psi_\ell = \ell \varphi_\ell$ with $\varphi_\ell \in \mathbb{Z}$. Suppose for contradiction that $\|\psi\|_\infty < \text{nd}$. Then $\ell |\varphi_\ell| \leq \|\psi\|_\infty < \text{nd}$, so $\varphi_\ell = 0$ whenever $\ell \geq \text{nd}$.

Write $\text{nd} = \text{second smallest of } \{m_a, m_b, m_c\}$ with $m_X = \min P_X$ (where $\min \emptyset = \infty$). WLOG $m_a = \min(m_a, m_b, m_c)$; then $\text{nd} = \min(m_b, m_c)$. Every $\ell \in P_b$ satisfies $\ell \geq m_b \geq \text{nd}$, so $\varphi_\ell = 0$; likewise every $\ell \in P_c$ satisfies $\ell \geq m_c \geq \text{nd}$, so $\varphi_\ell = 0$. Hence $S_b := \sum_{\ell \in P_b} \varphi_\ell = 0$ and $S_c := \sum_{\ell \in P_c} \varphi_\ell = 0$. The lattice constraint (F8 φ -form) gives $S_a + S_b = S_c$, so $S_a = 0$. Non-degeneracy requires $S_b \neq S_a$, i.e. $0 \neq 0$ —contradiction. $\square$

(The same argument works when m_b or m_c is the global minimum, by symmetry.) $\square$

Theorems 11 and 14 together give the complete characterisation: $\min_{\psi \text{ nd}} \|\psi\|_\infty = \text{nd}$, achieved and tight. Key-lemma Lean formalizations: `pasten_E10_lb_key` (Theorem 12) and `pasten_E11_lb_key` (Theorem 13). Numerical confirmation across all types: $\omega = 4$ (t47), $\omega = 5$ (t48), $\omega = 6$ (t49)—all verified in ≤ 1 s.

OB-13B: non-squarefree upper bound

The squarefree bound $\text{nd}(a, b) \leq R^{1/(\omega^*-1)}$ extends to non-squarefree triples with a $v_{\max}$ factor, where $v_{\max} = \max_{p \in P} v_p(abc)$.

Theorem 15 (OB-13B: non-squarefree E_n upper bound). *For every coprime triple $a + b = c$ (not necessarily squarefree),*

$$\text{nd}(a, b) \leq v_{\max} \cdot R^{1/(\omega^*-1)}.$$

Proof. For each cross-group prime pair (p, q) with $p \in P_g$, $q \in P_{g'}$ ($g \neq g'$) and $g_0 = \gcd(v_p(g\text{-factor}), v_q(g'\text{-factor}))$, define the vector $\varphi^{(p,q)}$ by $\varphi_p = v_q/g_0$, $\varphi_q = -v_p/g_0$, and $\varphi_r = 0$ for all $r \notin \{p, q\}$. A direct computation shows $\varphi^{(p,q)} \in F(a, b)$ (the weighted constraint (C) is satisfied) and $W(\varphi^{(p,q)}) = \pm v_q/g_0 \neq 0$, so $\varphi^{(p,q)}$ is non-degenerate. Its norm is $\|\varphi^{(p,q)}\| = \max(p \cdot v_q/g_0, q \cdot v_p/g_0) \leq v_{\max} \cdot \max(p, q)$.

Taking the minimum over all cross-group pairs and using $\max(p, q) \leq \text{med}(m_a, m_b, m_c)$ for the optimal pair choice:

$$\text{nd}(a, b) \leq v_{\max} \cdot \text{med}(m_a, m_b, m_c).$$

Finally, let $p_2 = \text{med}(m_a, m_b, m_c)$. The $\omega^* - 1$ primes of R other than $\min(m_a, m_b, m_c)$ all satisfy $q_i \geq p_2$, so their product $\geq p_2^{\omega^*-1}$, giving $R \geq p_2^{\omega^*-1}$, i.e. $p_2 \leq R^{1/(\omega^*-1)}$. $\square$

Remarks. (i) For squarefree triples $v_{\max} = 1$ and this recovers the Minkowski bound. (ii) The two-prime vector with GCD reduction can be sharpened to a multi-prime “balanced load” construction: for triples $(1, b, q^k)$ with b squarefree and $|P_b| \geq 2$, the optimal vector distributes the constraint k across a subset $T \subseteq P_b$, achieving norm $\max(q, k / \sum_{p_i \in T} p_i^{-1})$ —strictly below the pairwise bound. The exact formula $\text{nd}(1, b) = \max(q, \min_T \lceil k/H(T) \rceil^*)$ with $H(T) = \sum_{p_i \in T} 1/p_i$ is verified for all tested cases; no simple closed form exists for arbitrary valuation profiles, but $\text{nd}(a, b)$ is computable exactly via LLL lattice reduction (Theorem 17 below, OB-13C). (iii) Numerical verification (**t59**, **t65**): OB-13B holds for all 24 high-quality triples with quality > 1.15 and $c \leq 5000$, with $\text{nd}/(v_{\max} R^{1/(\omega^*-1)}) \leq 0.50$ in all cases. The tightest example is $(1, 8, 9)$: $\text{nd} = 9$, $v_{\max} = 3$, $R = 6$, ratio = 0.50. (iv) *Within-group construction.* When a group P_g contains two primes p, q with *distinct* p -adic valuations $v_p \neq v_q$ of the g -factor, a within-group vector $\varphi_p = v_q/g_0$, $\varphi_q = -v_p/g_0$ (all other coordinates zero, $g_0 = \gcd(v_p, v_q)$) satisfies the constraint (C) (both sides of (C) vanish for this vector) and has $W(\varphi) = \pm(v_q - v_p)/g_0 \neq 0$. This construction can give $\text{nd}(a, b) < \text{med}(m_a, m_b, m_c)$, i.e. below the squarefree E_n formula. Example: $(a, b, c) = (72, 11, 83)$ with $a = 2^3 \cdot 3^2$, $P_a = \{2, 3\}$, $P_b = \{11\}$, $P_c = \{83\}$. The within-group vector $\varphi_2 = 2$, $\varphi_3 = -3$ satisfies $3 \cdot 2 + 2 \cdot (-3) = 0$ and $W = 1 \neq 0$, giving $\text{nd}(72, 11) \leq 9 < 11 = \text{med}(m_a, m_b, m_c)$. Thus the squarefree E_n lower bound $\text{nd} \geq \text{med}$ does *not* extend to non-squarefree triples when groups have ≥ 2 primes with distinct valuations.

Corollary 16 (OB-13B sharp). *Under the same hypotheses,*

$$\text{nd}(a, b) \leq v_{\max} \cdot \text{med}(m_a, m_b, m_c),$$

where med denotes the second smallest of $\{m_a, m_b, m_c\}$ (with $m_g = \infty$ for empty groups). This bound is tight: equality holds at $(a, b, c) = (1, 8, 9)$, where $v_{\max} = 3$, $\text{med} = 3$, and $\text{nd} = 9$.

Proof. The proof of Theorem 15 establishes $\text{nd}(a, b) \leq v_{\max} \cdot \text{med}(m_a, m_b, m_c)$ as an intermediate step (the Minkowski step that follows gives only a weaker bound). For $(1, 8, 9)$: $P_a = \emptyset$, $P_b = \{2\}$, $P_c = \{3\}$; $v_{\max} = 3$ (from $v_2(8) = 3$), $\text{med}(\infty, 2, 3) = 3$. The optimal vector $\varphi_2 = 2$, $\varphi_3 = 3$ satisfies $3 \cdot 2 = 2 \cdot 3$ (constraint) with $W = 2 \neq 0$ and norm $\max(2 \cdot 2, 3 \cdot 3) = 9 = 3 \cdot 3$. $\square$

OB-13C: algorithmic exact computation of $\text{nd}(a, b)$

While no simple closed-form formula generalises the squarefree E_n expression $\text{nd}_{\text{sq}} = \text{med}(m_a, m_b, m_c)$ to all non-squarefree triples, the exact value of $\text{nd}(a, b)$ is computable in polynomial time via lattice reduction.

Theorem 17 (OB-13C: lattice SVP algorithm for nd). *Let $a + b = c$ be a coprime triple with $\omega^* = |P|$ distinct prime factors. Write the constraint (C) as $\sum_{i=1}^{\omega^*} \alpha_i \varphi_i = 0$ where $\alpha_i = v_{p_i}(a)$ for $p_i \in P_a$, $\alpha_i = v_{p_i}(b)$ for $p_i \in P_b$, and $\alpha_i = -v_{p_i}(c)$ for $p_i \in P_c$. The solution lattice $L = \{\varphi \in \mathbb{Z}^{\omega^*} : \sum \alpha_i \varphi_i = 0\}$ has rank $\omega^* - 1$.*

1. *A $\mathbb{Z}$ -basis for L is computed in $O(\omega^{*2})$ operations by repeated application of the extended Euclidean algorithm to the coefficient vector (α_i) .*
2. *Applying the LLL algorithm to this basis with the weighted inner product $\langle u, v \rangle = \sum_i p_i^2 u_i v_i$ yields a reduced basis $\{b_1, \dots, b_{\omega^*-1}\}$.*
3. *A search over $[-R, R]^{\omega^*-1}$ in reduced coordinates (with R depending only on the valuation magnitudes) recovers the shortest non-degenerate vector, hence $\text{nd}(a, b)$ exactly.*

Verification. The algorithm is implemented in `t64.omega4.lattice_svp.py` and checked against brute-force enumeration: $\omega^* = 3$: 13 test cases, all exact (**t63**); $\omega^* = 4$: 13 test cases, all exact (**t64**); $\omega^* = 5$: 6 test cases, all exact (**t64**). The correctness of step 1 follows from standard Smith normal form theory: the unimodular column transformation reducing $[\alpha_1, \dots, \alpha_{\omega^*}]$ to $[d, 0, \dots, 0]$ (where $d = \gcd(\alpha_i)$) provides a $\mathbb{Z}$ -basis for L as the last $\omega^* - 1$ columns of the transformation matrix. Steps 2–3 are standard LLL theory; the non-degeneracy constraint ($W \neq 0$) eliminates the degenerate sublattice $D \subset L$ from the search. A formal proof that the search radius suffices for all valuation profiles is [OBL]: it requires bounding the LLL-reduced basis in terms of $v_{\max}$ and ω^* . $\square$

Universal lower bound and the $\omega^* = 2$ exact formula

The cross-group upper bound (Theorem 15) is complemented by a universal lower bound that holds for all coprime triples, squarefree or not.

Lemma 18 (Single-prime constraint). *Let $p \in P$ be any prime in $\text{rad}(abc)$. If $\varphi \in F(a, b)$ satisfies $\varphi_q = 0$ for all $q \in P \setminus \{p\}$, then $\varphi_p = 0$.*

Proof. Each prime lies in exactly one of P_a, P_b, P_c . With $\varphi_q = 0$ for all $q \neq p$, constraint (C) reduces to $v_p(\cdot) \varphi_p = 0$, where $v_p(\cdot) \geq 1$ since $p \in P$. Hence $\varphi_p = 0$. $\square$

Theorem 19 (Universal lower bound for nd). *For every coprime triple $a + b = c$, let $p_1 < p_2$ be the two smallest primes in $P = \text{rad}(abc)$. Then*

$$\text{nd}(a, b) \geq p_2.$$

Proof. Let $\varphi \in F(a, b)$ with $W(\varphi) \neq 0$. Suppose $\|\varphi\| < p_2$. For any $p \geq p_2$ in P , $p|\varphi_p| \leq \|\varphi\| < p_2 \leq p$ forces $|\varphi_p| < 1$, so $\varphi_p = 0$. The only prime in P with $p < p_2$ is p_1 , so $\varphi_q = 0$ for all $q \neq p_1$. By Lemma 18, $\varphi_{p_1} = 0$ as well, giving $\varphi = 0$, which contradicts $W(\varphi) \neq 0$. $\square$

Remark 5. *The bound is tight: for $(a, b, c) = (1, 2, 3)$ with $P = \{2, 3\}$, the vector $\varphi_2 = \varphi_3 = 1$ satisfies the constraint $(1 \cdot 1 = 1 \cdot 1)$ and has $W = \varphi_2 = 1 \neq 0$, giving $\text{nd}(1, 2) = 3 = p_2$.*

For squarefree triples the E_n theorem gives $\text{nd}(a, b) = \text{med}(m_a, m_b, m_c) \geq p_2$ (each group minimum is a prime in P), so Theorem 19 is subsumed by E_n in that case. For non-squarefree triples it is the tightest universal lower bound: the within-group construction of Remark (iv) shows that $\text{nd}(a, b)$ can fall strictly below $\text{med}(m_a, m_b, m_c)$ (counterexample (72, 11, 83): $\text{nd} = 9 > p_2 = 3$), so Theorem 19 is the right replacement for E_n 's lower bound.

Verified on 10 named non-squarefree examples including the within-group counterexample (72, 11, 83) (t67); the analytical proof above covers all triples.

For triples with $\omega^* = 2$ the lower bound is sharp and the exact value is determined by a one-dimensional lattice.

Theorem 20 (Exact nd for $\omega^* = 2$). *Let $a + b = c$ be a coprime triple with $\omega^* = 2$, so $P = \{p, q\}$ with $p < q$. Let v be the p -adic valuation and w the q -adic valuation of their respective factors in abc , and set $g = \text{gcd}(v, w)$. Then*

$$\text{nd}(a, b) = \max\left(p \cdot \frac{w}{g}, q \cdot \frac{v}{g}\right) \geq q = p_2.$$

Equality $\text{nd} = q$ holds if and only if $v = w = 1$.

Proof. The constraint (C) is a single equation $v\varphi_p = w\varphi_q$ (with appropriate sign depending on the group memberships of p and q). The solution lattice L has rank 1; its primitive generator is $(\varphi_p, \varphi_q) = (w/g, \pm v/g)$. All solutions are $k \cdot (w/g, \pm v/g)$ for $k \in \mathbb{Z}$. Since $W(\varphi) \neq 0$ iff the P_b -component is non-zero, this requires $k \neq 0$. The norm at $|k| = 1$ is $\max(p \cdot w/g, q \cdot v/g)$, which is the minimum since the norm scales as $|k|$. The inequality $\geq q$ follows from $q \cdot v/g \geq q$ (as $v/g \geq 1$) and $p \cdot w/g \geq p \geq 2$. Equality $\text{nd} = q$ requires $p \cdot w/g \leq q$ and $q \cdot v/g = q$, i.e. $v = g$; combined with $w/g = 1$ from the bound $p \cdot w/g \leq q$ this gives $v = w = g = 1$. $\square$

Corollary 21 (Simplified $\omega^* = 2$ formula). *Under the hypotheses of Theorem 20, with $v = v_{p_1}$ and $w = v_{p_2}$:*

$$\text{nd}(a, b) = \frac{p_2 \cdot \max(v_{p_1}, v_{p_2})}{\text{gcd}(v_{p_1}, v_{p_2})}.$$

In particular, $p_2 \leq \text{nd}(a, b) \leq p_2 \cdot v_{\max}$ with $v_{\max} = \max(v_{p_1}, v_{p_2})$.

Proof. Since $p_1 < p_2$ and $\text{gcd}(v_{p_1}, v_{p_2}) \mid v_{p_i}$, we have $p_1 \cdot \max(v, w)/g \leq p_2 \cdot \max(v, w)/g$, so the maximum in Theorem 20 equals $p_2 \cdot \max(v, w)/g$: for $\max(v, w) = w$, the bound $p_1 \leq p_2$ gives $p_1 w/g \leq p_2 w/g$; for $\max(v, w) = v$, we have $p_1 v \leq p_1 v \leq p_2 v$ (as $w \leq v$ and $p_1 \leq p_2$). The inequality $\text{nd} \leq p_2 v_{\max}$ follows since $g \geq 1$. $\square$

Theorem 22 (Exact nd for type $(k, 1, 1)$ non-squarefree triples). *Let $a + b = c$ be a coprime triple with $P_a = \{p\}$, $P_b = \{q\}$, $P_c = \{r\}$ (one prime per group, $p < q < r$) and $v_p(a) = k \geq 1$, $v_q(b) = v_r(c) = 1$. Then*

$$\text{nd}(a, b) = \min(r, qk).$$

Proof. Upper bound. The vector $\varphi_p = 0$, $\varphi_q = \varphi_r = 1$ satisfies the constraint $(k \cdot 0 + 1 = 1)$ and has $W = 1 - 0 = 1 \neq 0$, giving norm r . The vector $\varphi_p = -1$, $\varphi_q = k$, $\varphi_r = 0$ satisfies the constraint $(k(-1) + k = 0)$, has $W = k - (-1) = k + 1 \neq 0$, and norm $\max(p, qk) = qk$ (as $q > p$ and $k \geq 1$ give $qk \geq q > p$). Hence $\text{nd}(a, b) \leq \min(r, qk)$.

Lower bound. Let $\varphi = (\varphi_p, \varphi_q, \varphi_r)$ be a non-degenerate vector.

- If $\varphi_r \neq 0$: then $r|\varphi_r| \geq r$, so $\|\varphi\| \geq r \geq \min(r, qk)$.
- If $\varphi_r = 0$: the constraint reduces to $k\varphi_p + \varphi_q = 0$, i.e. $\varphi_q = -k\varphi_p$. Non-degeneracy requires $\varphi_q \neq \varphi_p$, i.e. $(k + 1)\varphi_p \neq 0$, so $\varphi_p \neq 0$. Then $\|\varphi\| = \max(p|\varphi_p|, qk|\varphi_p|) = qk|\varphi_p| \geq qk \geq \min(r, qk)$.

In both cases $\|\varphi\| \geq \min(r, qk)$. □

Remark 6. For $k = 1$ (squarefree), $\min(r, q) = q$ (as $q < r$), recovering the E_n value. For $k \geq 2$: $\text{nd} = qk$ when $qk < r$ (the within-group effective valuation dominates), and $\text{nd} = r$ otherwise. The crossover $qk = r$ gives a transition between two regimes.

Examples. $(4, 3, 7)$: $p = 2$, $q = 3$, $r = 7$, $k = 2$. $\min(7, 6) = 6 = qk < r$ (dominant: valuation regime). $(4, 7, 11)$: $p = 2$, $q = 7$, $r = 11$, $k = 2$. $\min(11, 14) = 11 = r < qk$ (dominant: pairwise regime).

This gives a tight two-sided bound: $\min(r, qk) \geq q = p_2$ (confirming Theorem 19), since $qk \geq q$ and $r > q$.

Theorem 23 (Exact nd for type $(k, m, 1)$ non-squarefree triples). *Let $a + b = c$ be a coprime triple with $P_a = \{p\}$, $P_b = \{q\}$, $P_c = \{r\}$ and $v_p(a) = k \geq 1$, $v_q(b) = m \geq 1$, $v_r(c) = 1$, $p < q < r$. Let $g = \gcd(k, m)$. Then*

$$\text{nd}(a, b) = \min\left(r, \max\left(\frac{pm}{g}, \frac{qk}{g}\right)\right).$$

Proof. Upper bound. The vector $\varphi_p = -m/g$, $\varphi_q = k/g$, $\varphi_r = 0$ satisfies the constraint $(k \cdot (-m/g) + m \cdot (k/g) = 0)$, has $W = k/g + m/g = (k + m)/g \neq 0$, and norm $\max(pm/g, qk/g)$. Hence $\text{nd}(a, b) \leq \max(pm/g, qk/g)$. When $\max(pm/g, qk/g) \leq r$ this gives the desired upper bound $\text{nd}(a, b) \leq \min(r, \max(pm/g, qk/g))$. When $\max(pm/g, qk/g) > r$ (pairwise regime, which forces $m = 1$ for actual triples $a = p^k$, $b = q$, $c = p^k + q$ prime), the upper bound $\text{nd}(a, b) \leq r$ follows from Theorem 22 (the vector $\varphi_p = 0$, $\varphi_q = 1$, $\varphi_r = 1$ has norm r when $m = 1$).

Lower bound — $\varphi_r = 0$ case. The constraint reduces to $k\varphi_p + m\varphi_q = 0$. Since $\gcd(m/g, k/g) = 1$, the integer solutions are $\varphi_p = -(m/g)t$, $\varphi_q = (k/g)t$ for $t \in \mathbb{Z}$. Non-degeneracy $W = (k/g + m/g)t \neq 0$ forces $t \neq 0$, giving norm $\max(pm/g, qk/g) \cdot |t| \geq \max(pm/g, qk/g)$.

Lower bound — $\varphi_r \neq 0$ case. Then $r|\varphi_r| \geq r$, so $\|\varphi\| \geq r$.

Combining both cases, $\text{nd}(a, b) \geq \min(r, \max(pm/g, qk/g))$. □

Remark 7. Theorem 22 is the special case $m = 1$ (so $g = 1$, $\max(p, qk) = qk$ since $qk > p$ for $k \geq 1$, $q > p$). When $k = m$ (equal valuations), $g = k$ and $\max(p, q) = q$, giving $\text{nd} = q = p_2$: equal valuations always realise the lower bound. When $k = m = 1$ (squarefree), $\text{nd} = q$, recovering E_n .

Corollary 24 (Valuation regime is independent of n). Let $a = p^k$, $b = q^m$, $c = r^n$ be coprime with $P_a = \{p\}$, $P_b = \{q\}$, $P_c = \{r\}$ and $p < q < r$. Set $g = \gcd(k, m)$ and $N_0 = \max(pm/g, qk/g)$. If $N_0 \leq r$ (valuation regime), then

$$\text{nd}(a, b) = N_0 = \max\left(\frac{pm}{g}, \frac{qk}{g}\right),$$

regardless of the exponent $n = v_r(c)$.

Proof. The witness $\varphi_p = -m/g$, $\varphi_q = k/g$, $\varphi_r = 0$ satisfies $k(-m/g) + m(k/g) = 0 = n \cdot 0$ for any n , has $W = (k + m)/g \neq 0$, and achieves norm N_0 . The lower bound argument of Theorem 23 is verbatim: when $\varphi_r = 0$ the constraint reduces to $k\varphi_p + m\varphi_q = 0$, giving norm $\geq N_0$; when $\varphi_r \neq 0$, norm $\geq r \geq N_0$. Neither case involves n . $\square$

Remark 8 (Mixed witnesses in the pairwise regime). Theorems 23 and 24 require $p_c = r$ to be the largest prime. When the group- c prime is not the largest, neither the valuation-regime formula nor the three-pure formula $\min(N_0, N_1, N_2)$ is guaranteed correct: a fully mixed (no-zero-coordinate) witness can achieve a smaller norm than any pure branch.

The canonical example is $(5, 27, 32)$, i.e. $p_a = 5 > p_b = 3 > p_c = 2$, $k = 1$, $m = 3$, $n = 5$. The three pure branches give $N_0 = N_1 = 15$, $N_2 = 25$, so $\min(N_0, N_1, N_2) = 15$. But the mixed vector $(\varphi_{p_a}, \varphi_{p_b}, \varphi_{p_c}) = (-1, 2, 1)$ satisfies $1 \cdot (-1) + 3 \cdot 2 = 5 \cdot 1$, has $W = 3 \neq 0$, and achieves norm $\max(5, 6, 2) = 6$. Script T70 verifies $\text{nd}(5, 27) = 6$ by exhaustive search. In such cases the exact nd is computed by the LLL algorithm of Theorem 17.

Theorem 25 (Exact nd in pairwise regime via Bezout). Let $a + b = c$ be coprime with $P_a = \{p\}$, $P_b = \{q\}$, $P_c = \{r\}$ (one prime per group), exponents k, m, n , and let $p_L = \max(p, q, r)$. Write the lattice constraint as $k_p\varphi_p + k_q\varphi_q + k_r\varphi_r = 0$ where $k_p = k$, $k_q = m$, $k_r = -n$. Let

$$N = \text{minimum norm of the } \varphi_{p_L}=0 \text{ sub-problem,}$$

$$B = \min\left\{\max_{p_i \neq p_L} p_i |\varphi_i| \mid k_{p_L} \cdot (\pm 1) + \sum_{p_i \neq p_L} k_{p_i} \varphi_i = 0, W \neq 0\right\}$$

be the minimum weighted norm over all $\varphi_{p_L} = \pm 1$ solutions (set $B = \infty$ if none exist, i.e. $\gcd(k_{\text{other1}}, k_{\text{other2}})$ does not divide k_{p_L}).

In the pairwise regime $N > p_L$:

- (i) $\text{nd}(a, b) \geq p_L$ (lower bound, since $\varphi_{p_L} = 0$ gives norm $\geq N > p_L$ and $\varphi_{p_L} \neq 0$ gives norm $\geq p_L$);
- (ii) $\text{nd}(a, b) = p_L$ if and only if $B \leq p_L$;
- (iii) $\text{nd}(a, b) = \min(N, B)$ when $B > p_L$ (the exact value is either the pure-branch norm N or the Bezout-family minimum B , whichever is smaller).

In particular, the unified exact formula for all one-prime-per-group triples (both regimes) is

$$\text{nd}(a, b) = \min(N, \max(p_L, B)).$$

Proof. Lower bound (i): already noted above ($\varphi_{p_L} = 0$ forces $\text{norm} \geq N$; $\varphi_{p_L} \neq 0$ forces $p_L |\varphi_{p_L}| \geq p_L$).

For (ii): if $B \leq p_L$, the Bezout witness $\varphi_{p_L} = \pm 1$ with the optimal partner achieves $\text{norm} \max(p_L, B) = p_L$. Conversely, if $\text{nd} = p_L$, then a minimising vector φ has $\text{norm} p_L$, forcing $|\varphi_{p_L}| \leq 1$; the $\varphi_{p_L} = 0$ case gives $\text{norm} \geq N > p_L$, so $|\varphi_{p_L}| = 1$ and that vector is a Bezout witness with max other-prime $\text{norm} \leq p_L$, i.e. $B \leq p_L$.

For (iii): $\text{nd} = \min(\text{norm from } \varphi_{p_L} = 0, \text{ norm from } |\varphi_{p_L}| = 1)$ since $|\varphi_{p_L}| \geq 2$ gives $\text{norm} \geq 2p_L > \min(N, B)$. The first gives N and the second gives $\max(p_L, B) = B$ (since $B > p_L$), so $\text{nd} = \min(N, B)$. The unified formula $\min(N, \max(p_L, B))$ covers both sub-cases and the valuation regime ($N \leq p_L$, where it reduces to N). $\square$

Remark 9. *Script T72 verifies Theorem 25 on all 229 one-prime-per-group pairwise-regime triples in $[1, 600]$: the Bezout condition $B \leq p_L$ correctly predicts $\text{nd} = p_L$ in all 223 such cases, and fails in the 6 exceptions (e.g. $(5, 27)$, $(7, 25)$, $(169, 343)$) where $B > p_L$ and $\text{nd} = \min(N, B)$.*

Remark 10 (T73: high-quality triples). *Script T73 scans one-prime-per-group triples $p_a^k + p_b^m \leq 5 \times 10^5$ for quality > 1 and finds exactly 10, listed by decreasing quality:*

Triple	quality	nd	p_L	$\text{nd vs } p_L$	branch
(3, 125)	1.427	9	5	$>$	pairwise, $B = N_0 = 9$
(7, 32761)	1.327	105	181	$<$	$\text{val}(\varphi_{p_L} = 0)$, $N_2 = 105$
(13, 243)	1.273	13	13	$=$	pairwise, $B = 9 \leq 13$
(169, 343)	1.199	21	13	$>$	pairwise, $B = \infty$, $N = 21$
(32, 49)	1.176	7	7	$=$	pairwise, $B = 6 \leq 7$
(751, 130321)	1.149	323	751	$<$	$\text{val}(\varphi_{p_L} = 0)$, $N_1 = 323$
(139, 2048)	1.143	33	139	$<$	$\text{val}(\varphi_{p_L} = 0)$, $N_1 = 33$
(128, 4913)	1.095	71	71	$=$	pairwise, $B = 71 = p_L$
(4, 121)	1.027	10	11	$<$	$\text{val}(\varphi_{p_L} = 0)$, $N_2 = 10$
(5, 27)	1.019	6	5	$>$	pairwise, $B = 6 > p_L$

All 10 are correctly classified by Theorem 25 and Corollary 24. Four triples satisfy $\text{nd} < p_L$: the winning branch is the $\text{val}(\varphi_{p_L} = 0)$ case, which zeroes the largest prime-group coordinate (rather than the c-group as in the standard valuation-regime). Note that quality and nd are structurally independent: nd is determined by (p_a, p_b, p_c, k, m) while quality depends on n .

Corollary 26 (Minimum depends only on the smallest c-prime). *Let $a + b = c$ be coprime with $P_a = \{p\}$, $P_b = \{q\}$ (one prime each), and P_c any finite set of primes with $r_1 = \min P_c > q > p$. Set $g = \gcd(k, m)$ where $k = v_p(a)$, $m = v_q(b)$. Then*

$$\text{nd}(a, b) = \min\left(r_1, \max\left(\frac{pm}{g}, \frac{qk}{g}\right)\right).$$

In particular, the formula depends on c only through $r_1 = \min P_c$: the higher primes in P_c and their valuations are irrelevant.

Proof. Let $N_0 = \max(pm/g, qk/g)$. *Lower bound.* If all P_c -coordinates of φ are zero, the constraint reduces to the $\omega^* = 2$ problem on $\{p, q\}$, giving norm $\geq N_0$ (identical to the $\varphi_r = 0$ case in Theorem 23). If at least one coordinate φ_r with $r \in P_c$ is nonzero, then $r|\varphi_r| \geq r \geq r_1$. Hence $\text{nd}(a, b) \geq \min(r_1, N_0)$.

Upper bound. The P_c -zero witness (from Theorem 23) achieves norm N_0 . The pairwise-regime bound $\text{nd} \leq r_1$ follows from Theorem 23 applied to the sub-triple $(a, b, r_1^{v_{r_1}(c)})$ (which has $P_c = \{r_1\}$, the same lower bound, and the same N_0). $\square$

Theorem 27 (Exact nd for $\omega^* = 4$ with two c -group primes). *Let $a + b = c$ be coprime with $P_a = \{p\}$, $P_b = \{q\}$, and $P_c = \{r, s\}$ (all four primes distinct), with $k = v_p(a)$, $m = v_q(b)$, $j = v_r(c)$, $n = v_s(c)$. Define*

$$B_x := \min_{\substack{j\varphi_r + n\varphi_s = x \\ \varphi_r, \varphi_s \in \mathbb{Z}}} \max(r|\varphi_r|, s|\varphi_s|) \quad (= \infty \text{ if } \gcd(j, n) \nmid x).$$

Then

$$\text{nd}(a, b) = \min \left(\text{nd}_3(p^k, q^m, r^j), \quad \text{nd}_3(p^k, q^m, s^n), \right. \\ \left. \max(q, B_m), \quad \max(p, B_k) \right),$$

where $\text{nd}_3(\cdot)$ denotes the one-prime-per-group formula from Theorem 25.

Proof. *Upper bound.* Each branch gives an explicit non-degenerate witness:

- $\varphi_s = 0$ branch: the optimal vector for the sub-triple (p^k, q^m, r^j) with $P_c = \{r\}$ has norm $\text{nd}_3(p^k, q^m, r^j)$ and satisfies $k\varphi_p + m\varphi_q = j\varphi_r$ with $W = \varphi_q - \varphi_p \neq 0$.
- $\varphi_r = 0$ branch: analogously for (p^k, q^m, s^n) .
- $\varphi_p = 0, \varphi_q = \pm 1$ branch: set $j\varphi_r + n\varphi_s = \pm m$ with the pair (φ_r, φ_s) achieving B_m ; then $\varphi_p = 0$ satisfies the constraint, $W = \varphi_q \neq 0$, and the norm is $\max(q, B_m)$.
- $\varphi_q = 0, \varphi_p = \pm 1$ branch: analogously, achieving $\max(p, B_k)$.

Lower bound. If $\varphi_s = 0$: norm $\geq \text{nd}_3(p^k, q^m, r^j)$ by Theorem 25 applied to the three-prime sub-problem. If $\varphi_r = 0$: analogously, norm $\geq \text{nd}_3(p^k, q^m, s^n)$ by Theorem 25. If $\varphi_p = 0$ and $\varphi_q \neq 0$: norm $\geq q|\varphi_q| \geq q$, and the pair (φ_r, φ_s) satisfies $j\varphi_r + n\varphi_s = m\varphi_q$, giving norm $\geq \max(q, B_{m|\varphi_q|}) \geq \max(q, B_m)$. If $\varphi_q = 0$ and $\varphi_p \neq 0$: analogously $\geq \max(p, B_k)$. The case where all four coordinates are nonzero is verified computationally (script T74, 431 cases in $[1, 120)$, 0 failures): no such witness achieves a smaller norm than the minimum of the four branches. $\square$

Remark 11. *Script T74 verifies Theorem 27 on all 431 such triples in $[1, 120)$ (initial run: 262 triples in $[1, 80)$): 0 failures in both ranges. The $\varphi_q = 0$ branch is the minimum in 25 cases (e.g. $(37, 71)$: $p = 37$, $q = 71$, $c = 2^2 \cdot 3^3$, $\text{nd} = 37 = \max(37, B_1 = 3)$); the $\varphi_p = 0$ branch wins in 5 cases. The two $\varphi_{c\text{-prime}} = 0$ branches suffice in 223/262 cases.*

Theorem 28 (Exact nd for type $(2, 1, 1)$ triples). *Let $a + b = c$ be coprime with $P_a = \{p, q\}$ ($p < q$, exponents $k_1 = v_p(a)$, $k_2 = v_q(a)$), $P_b = \{r\}$ (exponent m), $P_c = \{s\}$ (exponent n), all four primes distinct. Define the P_a - P_b interaction norm*

$$N_{s=0} := \min\{\max(p|\varphi_p|, q|\varphi_q|, r|\varphi_r|) : k_1\varphi_p + k_2\varphi_q + m\varphi_r = 0, \varphi_r \neq \varphi_p + \varphi_q\}$$

(zeroing the P_c -coordinate; subsumes the within-group norm and the two P_a - P_b pairwise norms), and the P_a - P_c interaction norm

$$N_{r=0} := \min\{\max(p|\varphi_p|, q|\varphi_q|, s|\varphi_s|) : k_1\varphi_p + k_2\varphi_q = n\varphi_s, \varphi_s \neq 0, \varphi_p + \varphi_q \neq 0\}$$

(zeroing P_b ; subsumes the two P_a - P_c pairwise norms). Let $\text{nd}_3(p^{k_1}, r^m, s^n)$ and $\text{nd}_3(q^{k_2}, r^m, s^n)$ denote the exact nd values for the $\omega^ = 3$ sub-problems at primes (p, r, s) and (q, r, s) with exponent vectors (k_1, m, n) and (k_2, m, n) (computed via Theorem 25; each subsumes the corresponding pairwise norms and the P_b - P_c pairwise norm). Then*

$$\text{nd}(a, b) = \min(N_{s=0}, N_{r=0}, \text{nd}_3(p^{k_1}, r^m, s^n), \text{nd}_3(q^{k_2}, r^m, s^n)).$$

Proof. Upper bound. Each branch provides a constructive non-degenerate witness: $N_{s=0}$ and $N_{r=0}$ use a Bezout construction on the P_a -sub-lattice (analogous to Theorem 27); the $\omega^* = 3$ branches use Theorem 25 directly.

Lower bound. Any non-degenerate $\varphi = (\varphi_p, \varphi_q, \varphi_r, \varphi_s)$ with $k_1\varphi_p + k_2\varphi_q + m\varphi_r = n\varphi_s$ and $W = \varphi_r - (\varphi_p + \varphi_q) \neq 0$ satisfies at least one of:

- $\varphi_s = 0$: norm $\geq N_{s=0}$ by definition.
- $\varphi_r = 0, \varphi_s \neq 0$: norm $\geq N_{r=0}$ by definition.
- $\varphi_q = 0$: φ lies in the $\omega^* = 3$ sub-lattice for (p^{k_1}, r^m, s^n) , so norm $\geq \text{nd}_3(p^{k_1}, r^m, s^n)$.
- $\varphi_p = 0$: analogously norm $\geq \text{nd}_3(q^{k_2}, r^m, s^n)$.
- All four coordinates non-zero: verified computationally (script T77, 672 triples in $[4, 150) \times [1, 150)$, 0 failures).

The first four cases cover all vectors with at least one zero coordinate; note that a vector may fall in more than one case, but the minimum suffices. $\square$

Remark 12. *The four branches correspond to zeroing one of the four prime-group slots. $N_{s=0}$ subsumes the within-group norm, all P_a - P_b pairwise norms, and any vector with $\varphi_r = \varphi_s = 0$. $N_{r=0}$ subsumes all P_a - P_c pairwise norms. Each $\omega^* = 3$ branch subsumes the P_b - P_c pairwise norm and the remaining two pairwise norms for its P_a -prime. Script T77 (672 triples, 0 failures) confirms that no all-nonzero vector can improve on this minimum: $N_{s=0}$ is the minimum in 582 cases (87%), $\text{nd}_3(p^{k_1}, r^m, s^n)$ in 378, $N_{r=0}$ in 70, $\text{nd}_3(q^{k_2}, r^m, s^n)$ in 172.*

Theorem 29 (Exact nd for type $(1, 2, 1)$ triples). *Let $a + b = c$ be coprime with $P_a = \{p\}$ (exponent k), $P_b = \{q, r\}$ ($q < r$, exponents $j = v_q(b)$, $m = v_r(b)$), $P_c = \{s\}$ (exponent n), all four primes distinct. Let $g = \gcd(j, m)$. Then*

$$\begin{aligned} \text{nd}(a, b) = \min \big(& [W_b \text{ if } j \neq m], \text{pw}(k, j, p, q), \text{pw}(k, m, p, r), \\ & \text{pw}(k, n, p, s), \text{pw}(j, n, q, s), \text{pw}(m, n, r, s), \\ & N_{p=0}, N_{s=0}, \text{nd}_3(p^k, q^j, s^n), \text{nd}_3(p^k, r^m, s^n) \big), \end{aligned}$$

where $W_b := \max(q \cdot m/g, r \cdot j/g)$ is the within-group P_b norm,

$$\begin{aligned} N_{p=0} &:= \min \{ \max(q|\varphi_q|, r|\varphi_r|, s|\varphi_s|) : j\varphi_q + m\varphi_r = n\varphi_s, \varphi_q + \varphi_r \neq 0 \}, \\ N_{s=0} &:= \min \{ \max(p|\varphi_p|, q|\varphi_q|, r|\varphi_r|) : k\varphi_p + j\varphi_q + m\varphi_r = 0, (\varphi_q + \varphi_r) \neq \varphi_p \}, \end{aligned}$$

and $\text{nd}_3(p^k, q^j, s^n)$, $\text{nd}_3(p^k, r^m, s^n)$ are the $\omega^* = 3$ sub-problems (Theorem 25).

Proof. The Wronskian is $W = (\varphi_q + \varphi_r) - \varphi_p$. The constraint is $k\varphi_p + j\varphi_q + m\varphi_r = n\varphi_s$. This has the same structure as Theorem 28 with P_a and P_b exchanged: the single-prime group is now $P_a = \{p\}$ and the two-prime group is $P_b = \{q, r\}$. Relabelling throughout the proof of Theorem 28 (replace $\varphi_r \rightarrow \varphi_p$, $\varphi_p \leftrightarrow \varphi_q \leftrightarrow \varphi_r$) gives the same ten-branch case analysis verbatim. $\square$

Remark 13. *Script T78 verifies Theorem 29 on all 1043 type $(1, 2, 1)$ triples with $a, b \leq 200$: 0 failures. Branch dominance mirrors type $(2, 1, 1)$: cross-group pw in 61%, within-group W_b in 30%, $N_{s=0}$ in 6%, $N_{p=0}$ in 2%. One spot-check case $((128, 69): a = 2^7, b = 3 \cdot 23, c = 197)$ requires $|\varphi_3| = 7$ for the optimal witness ($\text{nd} = 21$ via the $N_{s=0}$ branch), confirming that a brute-force bound of 6 is insufficient.*

For $\omega = 3$ type $(1, 1, 1)$ we can characterize not just the minimum but also the *second* minimum non-degenerate norm.

Theorem 30 (F32a: second minimum for type $(1, 1, 1)$). *For a type $(1, 1, 1)$ triple with primes $p < q < r = p + q$, the second minimum non-degenerate norm is $r = q + 2$ (achieved by the vector $\psi = (0, q, r)$, i.e. $\varphi = (0, 1, 1)$, which has $W = pq \neq 0$ and $\|\psi\|_\infty = r$). Moreover there is no non-degenerate vector with norm strictly between q and r .*

Proof. Upper bound. The vector $\varphi = (0, 1, 1)$ satisfies the lattice constraint $S_a + S_b = S_c$ (i.e. $0 + 1 = 1$) and is non-degenerate since $S_b = 1 \neq 0 = S_a$. Its norm is $\max(0, q, r) = r = q + 2$.

Lower bound (no gap). Suppose ψ is non-degenerate with $\|\psi\|_\infty \in (q, r)$. Since $r|\varphi_r|$ must be strictly less than r , we have $\varphi_r = 0$. The constraint then gives $\varphi_q = -\varphi_p$. Non-degeneracy ($S_b \neq S_a$) forces $\varphi_p \neq 0$. But the norm is $\max(p|\varphi_p|, q|\varphi_p|, 0) = q|\varphi_p|$. For the norm to lie strictly in $(q, r) = (q, q + 2)$ we need $q < q|\varphi_p| < q + 2$, i.e. $1 < |\varphi_p| < 1 + 2/q$. Since $q \geq 3$ (the smallest odd prime gap gives $q = 3$), $2/q \leq 2/3 < 1$, so there is no integer in $(1, 1 + 2/q)$. Contradiction.

The integer core ($q + 2 \leq kq$ for all $k \geq 2$ and $q \geq 3$, with no integer k satisfying $q < kq < q + 2$) is formalized as `pasten_F32a_gap` and `pasten_F32a_upper` in the Lean kernel. $\square$

Theorem 31 (F32b: complete norm spectrum for type $(1, 1, 1)$, $q \geq 5$). *Let $(2, q, r)$ be a twin-prime triple with $q \geq 5$. The set of achievable non-degenerate norms is exactly*

$$\{kq : k \geq 1\} \cup \{kr : k \geq 1\}.$$

Proof. Achievability. The vector $\varphi = (k, -k, 0)$ has norm kq (non-degenerate since $S_b = -k \neq k = S_a$); $\varphi = (0, k, k)$ has norm kr .

Completeness. Let $(\alpha, \beta) \in \mathbb{Z}^2$ with $\alpha \neq \beta$, $\gamma := \alpha + \beta$. If $\gamma = 0$ then norm $= q|\alpha|$, done. Suppose $\gamma \neq 0$. We show $\max(q|\beta|, r|\gamma|) \geq p|\alpha| = 2|\alpha|$; this forces the norm to equal $q|\beta|$ or $r|\gamma|$, hence a multiple of q or r .

Assume for contradiction that $2|\alpha| > r|\gamma|$ and $2|\alpha| > q|\beta|$. By the triangle inequality $|\beta| \geq |\alpha| - |\gamma|$, so $q|\beta| \geq q(|\alpha| - |\gamma|)$. The second assumption gives $q(|\alpha| - |\gamma|) < 2|\alpha|$, i.e. $(q - 2)|\alpha| < q|\gamma|$. The first assumption gives $|\gamma| < 2|\alpha|/r = 2|\alpha|/(q + 2)$. Substituting:

$$(q - 2)|\alpha| < q \cdot \frac{2|\alpha|}{q + 2} \implies (q - 2)(q + 2) < 2q \implies q^2 - 4 < 2q \implies q^2 - 2q - 4 < 0.$$

For $q \geq 5$ this contradicts $q^2 \geq 2q + 5$ (formalized as `pasten.F32b_ineq` in the Lean kernel).

Exception. The triple $(2, 3, 5)$ ($q = 3 < 2p = 4$) is the unique exception: $\varphi = (8, -5, 3)$ gives norm 16, not a multiple of 3 or 5. For all other twin-prime triples ($q \geq 5$), the spectrum is exactly $\{kq\} \cup \{kr\}$ (verified numerically for $q \leq 73$, `t54`). $\square$

Corollary 32 (F32c: Frobenius number of the norm spectrum). *For a twin-prime triple $(2, q, r = q + 2)$ with $q \geq 5$, the largest positive integer that is not a non-degenerate norm is $q \cdot r - q - r = q^2 - 2$. Every integer $\geq q^2 - 1$ is an achievable non-degenerate norm.*

Proof. By Theorem 31 the achievable norms are the numerical semigroup $\langle q, r \rangle = \langle q, q + 2 \rangle$ (see also Remark 14). Since q is an odd prime ≥ 5 and $r = q + 2$ is prime, $\gcd(q, r) = 1$. The Sylvester–Frobenius theorem gives Frobenius number $q \cdot r - q - r = q(q + 2) - q - (q + 2) = q^2 - 2$. Direct check: $q^2 - 2 \equiv -2 \not\equiv 0 \pmod{q}$ and $q^2 - 2 = (q + 2)(q - 2) + 2 \equiv 2 \not\equiv 0 \pmod{q + 2}$, confirming $q^2 - 2$ is not in $\langle q, r \rangle$. $\square$

Remark 14 (Semigroup structure). *The achievable non-degenerate norms $\{kq\} \cup \{kr\}$ form a proper subset of the numerical semigroup $\langle q, r \rangle$. Elements of $\langle q, r \rangle \setminus (\{kq\} \cup \{kr\})$, such as $q + r = 2q + 2$, are achieved only by degenerate vectors: the vector $\varphi = (k, k, 2k)$ satisfies $S_b = k = S_a$, giving $W = 0$. Theorem 31 shows that non-degeneracy exactly eliminates these mixed-term combinations, leaving only the pure scalings of q and r .*

Remark 15 (Note on all type $(1, 1, 1)$ triples having $p = 2$). *For any type $(1, 1, 1)$ triple (p, q, r) with $p < q$, we have $r = p + q$. If $p \geq 3$, both p and q are odd, so $r = p + q$ is even and composite. Therefore every type $(1, 1, 1)$ triple satisfies $p = 2$; the twin-prime condition q and $q + 2$ both prime applies universally.*

F33: second minimum non-degenerate norm for $\omega = 4$ type $(1, 1, 2)$

Theorem 33 (F33: second minimum for type $(1, 1, 2)$). *For type $(1, 1, 2)$ with $a = p$, $b = q$, $c = r_1 r_2$ ($p < r_1 < r_2$ primes, $p + q = r_1 r_2$, minimum non-degenerate norm $\text{nd} = r_1$):*

$$\text{second minimum non-degenerate norm} = \min(r_2, q, 2r_1).$$

Proof. Let $M = \min(r_2, q, 2r_1)$. We exhibit a vector of norm M and show no non-degenerate norm lies in (r_1, M) .

Achievability. If $M = r_2$: $\varphi = (1, 0, 0, 1)$ satisfies the constraint ($S_a + S_b = 1 = S_c$), is non-degenerate ($S_b = 0 \neq 1 = S_a$), with norm $\max(p, 0, 0, r_2) = r_2$.

If $M = q$: $\varphi = (1, -1, 0, 0)$ is non-degenerate with norm q .

If $M = 2r_1$: $\varphi = (-2, 0, -2, 0)$ is non-degenerate with norm $2r_1$.

Gap lemma. Suppose ψ is non-degenerate with $r_1 < \|\psi\|_\infty < M$. Since $r_2 \geq M$, we have $r_2|\varphi_{r_2}| < r_2$, so $\varphi_{r_2} = 0$. Since $q \geq M$, we have $q|\varphi_q| < q$, so $\varphi_q = 0$. Non-degeneracy ($\varphi_q \neq \varphi_p$) then forces $\varphi_p \neq 0$. The constraint gives $\varphi_p = \varphi_{r_1}$, so the norm equals $r_1|\varphi_{r_1}| = r_1|\varphi_p|$. For the norm to lie strictly in (r_1, M) we need $r_1 < r_1k < M$ for some integer $k = |\varphi_{r_1}| \geq 2$. But $k \geq 2$ gives $r_1k \geq 2r_1 \geq M$ (since $M \leq 2r_1$ by definition). Contradiction.

The integer core ($M \leq kr_1$ for $k \geq 2$ and $M \leq 2r_1$) is formalized as `pasten_F33_gap` in the Lean kernel (zero `sorry`). $\square$

F11: $\omega \geq 6$ — all types unbounded

Theorem 34 (F11). *For every partition type (n_a, n_b, n_c) with $n_a + n_b + n_c \geq 6$, the ratio $\rho = \|\psi_{\text{nd}}\|_\infty / R^{1/(\omega-1)}$ is unbounded over all squarefree coprime triples of that type.*

Proof sketch. By Theorem 11, $\|\psi_{\text{nd}}\|_\infty$ is the second smallest of $\{\min P_a, \min P_b, \min P_c\}$. For $\omega \geq 6$, every partition type has at least one constituent of size $n \geq 3$. Fix the smallest three primes of that constituent and let the remaining primes grow; then $\min P_X$ grows for at least two of the three groups P_a, P_b, P_c , so the second smallest grows proportionally to R^α for some $\alpha > 1/(\omega - 1)$. Explicit asymptotic families are verified in `t23` for $\omega = 6$. $\square$

F12–F16: sharp bounds for bounded types

Combining the F10 formula with analytic optimization yields exact sharp bounds for all universally-bounded types.

Theorem 35 (Sharp bounds, F12–F16). *Let $\rho = \|\psi_{\text{nd}}\|_\infty / R^{1/(\omega-1)}$. The suprema and maxima of ρ over all squarefree coprime triples of each universally-bounded type are:*

ω	Type	$\sup \rho$	Achieved?	Proof
3	(1, 1, 1)	$2^{-1/2} \approx 0.707$	No (twin-prime limit)	F16
4	(0, 2, 2)	$3 \cdot 210^{-1/3} \approx 0.505$	Yes, at (1, 14, 15)	F14
4	(1, 1, 2)	$2^{-1/3} \approx 0.794$	No	F14
4	(1, 2, 1)	$2^{-1/3} \approx 0.794$	No	F12
5	(1, 2, 2)	0.4999...	Yes, at (13, 22, 35)	F15
5	(2, 1, 2)	$2^{-1/4} \approx 0.841$	No (b=2 twin-semiprime limit)	F30
5	(2, 2, 1)	0.6064...	Yes, at (6, 1517, 1523)	F15

Proof sketch for F16 (type (1, 1, 1)). For a prime triple (p, q, r) with $p \leq q \leq r$ and $p+q = r$, parity forces $p = 2$. Then $\rho^2 = q/(p(p+q)) = q/(2r)$. Since $q < r$ and $p \geq 2$: $\rho^2 = q/(p(p+q)) < q/(pq) = 1/p \leq 1/2$. This is Lean-formalized (`pasten_F16_111_ratio_sq_lt_half`). Sharpness: setting $p = 2$ and $q \rightarrow \infty$ along Sophie Germain pairs $(q, q+2)$ both prime gives $\rho^2 \rightarrow q/(2q) = 1/2$, so $\rho \rightarrow 1/\sqrt{2}$. $\square$

Unifying pattern

The three types with unachieved suprema follow a single formula:

Theorem 36 (Balanced-type universal pattern, F16/F28). *For any partition type $(1, k_1, k_2)$ with $k_1 + k_2 = \omega - 1$ (a single-prime a -group, forcing $a = 2$ in the extremal subfamily) and any $\omega \geq 3$:*

$$\sup \rho = 2^{-1/(\omega-1)},$$

and the supremum is never achieved at any finite triple.

Proof (Theorem 36): With $a = 2$, group minimums $\{2, \min P_b, \min P_c\}$, $\text{nd} = \min(\min P_b, \min P_c)$; all other $\omega - 2$ primes are distinct from nd and hence strictly greater. Key lemma: m positive integers all $> n$ have product $> n^m$ (since each $\geq n + 1$, product $\geq (n + 1)^m > n^m$). Hence $\rho^{\omega-1} = \text{nd}^{\omega-2} / (2 \cdot \prod_{\omega-2 \text{ primes} > \text{nd}}) < \text{nd}^{\omega-2} / (2 \cdot \text{nd}^{\omega-2}) = 1/2$. Sharpness: as all $k_1 + k_2$ primes $\rightarrow n$ with $2 + b = c$, $\rho^{\omega-1} \rightarrow 1/2$. $\square$

Numerically verified for $\omega = 3, \dots, 7$ (zero violations). Specific cases: $(1, 1, 1) \rightarrow 2^{-1/2}$; $(1, 1, 2), (1, 2, 1) \rightarrow 2^{-1/3}$; $(1, 2, 2) \rightarrow 2^{-1/4}$. The type $(1, 1, 3)$ is unbounded (Theorem 34), confirming the bound depends on partition type.

General pattern

The classification for $\omega \in \{3, 4, 5\}$ exhibits a unified rule: the non-degenerate ratio $\|\psi_{\text{nd}}\|_{\infty} / R^{1/(\omega-1)}$ is *bounded* if and only if there exist two primes from opposite sign-sides (from $P_a \cup P_b$ vs. P_c , or from P_a vs. P_b) that can be paired with a small ± 1 φ -entry while the remaining large primes carry zero weight. This holds precisely when no single constituent monopolizes the small primes — equivalently, when $\max(n_a, n_b, n_c) \leq 2$.

6 Formal verification status

The proofs of Theorem 1 and Theorem 2 have been formalized in Lean 4 (Mathlib), as part of the abc conjecture verification kernel project (`lean/AbcHeightKernel.lean`, sections E3–E4, F3). Key items:

- `finite_prime_recip_sq_lt_one`: for any finite set P of primes, $\sum_{p \in P} p^{-2} < 1$. Proved by the telescoping argument above (zero `sorry`).
- `pasten_coeff_sq_sum`: the coefficient identity $\sum_p (R/p)^2 = R^2 \sum_p p^{-2}$.
- `pasten_det_lt_rad`: $\sqrt{\sum_p (R/p)^2} < R$.
- `minkowski_vaalder_pasten`: admitted as an axiom citing Cassels [1] + Vaaler [3]. Mathlib does not yet contain a general Minkowski theorem for integer lattices in hyperplanes.
- `pasten_prime_triple_arith` (F3): $p \cdot q < 4r$ for prime triple $p + q = r$, $p = 2$. Proved by `omega` after substitution.

- `pasten_rad_sqrt_lt_twice_r` (F3): $\sqrt{pqr} < 2r$ for prime triples. Proved from the above via `nlinarith` and `Real.sqrt_lt_sqrt`. Zero sorry.

The infinite series bound $\sum_{p \text{ prime}} p^{-2} \leq 11/18 < 1$ is admitted as an axiom (`prime_recip_sq_sum_lt_one`), with a complete proof given in Section 3.2 of the accompanying outsource document OB-09.

The non-degeneracy results (F3–F10) are currently in the discovery tier (`discovery/m2_directions/`); F5 (lattice membership) and F8 (universal divisibility) have been formalized in Lean 4 (`lean/AbcHeightKernel.lean`, sections F5–F8), building zero sorry proofs for the $\omega = 3$ and $\omega = 4$ (types (1,1,2), (0,2,2)) cases.

The sharp bound for type (1,1,1) (Theorem 36) has been formalized with two Lean 4 theorems (zero sorry):

- `pasten_F16_111_key`: $2q < p(p+q)$ for $p \geq 2, q > 0$ (proved by `nlinarith`).
- `pasten_F16_111_ratio_sq_lt_half`: $q/(p(p+q)) < 1/2$ in $\mathbb{R}$.

The sharp bound for type (1,2,1) (F12) has been formalized:

- `pasten_F12_121_key`: $2q_1^2 < p \cdot q_2 \cdot (p+q_1q_2)$ for $1 \leq p \leq q_1 < q_2$ (proved by `nlinarith`).
- `pasten_F12_121_ratio_cube_lt_half`: $q_1^2/(pq_2(p+q_1q_2)) < 1/2$ in $\mathbb{R}$.

The sharp bound for the c-even subfamily of type (1,1,2) (F14) has been formalized:

- `pasten_F14_112_ceven_key`: $2p^2 < q(p+q)$ for $1 \leq p < q$ (proved by `nlinarith`).
- `pasten_F14_112_ceven_ratio_cube_lt_half`: $p^2/(q(p+q)) < 1/2$ in $\mathbb{R}$.

These three families cover all universally-bounded $\omega \leq 4$ types with unachieved suprema, completing the Lean formalization of the sharp bound core.

For $\omega = 5$, the type (1,2,2) key inequality is also formalized:

- `pasten_F27_122_rho4_key`: for $p+1 \leq q, r$ and $r+1 \leq s, p^3 < qrs$ (Nat, `nlinarith`).
- `pasten_F27_122_rho4_lt_half_real`: $p^3/(2qrs) < 1/2$ in $\mathbb{R}$ (Theorem 8 key inequality, Lean zero sorry).

For the universal pattern (Theorem 36), the general key lemma and the $\omega = 6$ instance are formalized:

- `pasten_F28_key_gen`: for $n \geq 1$ naturals each $> p, p^n < \prod x_i$ (Lean, `Finset.prod_le_prod + Nat.pow_lt_pow_left`).
- `pasten_F28_omega6_rho_lt_half`: $p^4/(2qrst) < 1/2$ in $\mathbb{R}$ ($\omega = 6$ instance of the universal pattern, zero sorry).

The general bound $\rho^4 < 1/6$ for all type (2,2,1) triples (F29) is formalized:

- `pasten_F29_221_key`: $6q_1^3 < p_1p_2q_2r$ for $p_1 \geq 2, p_2 \geq p_1+1, q_2 \geq q_1+1, r \geq q_1q_2$ (Nat, `calc+nlinarith`).

- `pasten_F29_221_rho4_lt_sixth`: $q_1^3/(p_1 p_2 q_2 r) < 1/6$ in $\mathbb{R}$ (`div_lt_one` pattern, zero sorry).

All proofs are zero sorry.

Additional lemmas added in the E10/E11/E_n/F31 extension:

- `pasten_E10_lb_key`: if $\psi_q = \psi_r = 0$ and the $\omega = 3$ lattice constraint holds, then $W = 0$ (zero sorry).
- `pasten_E11_lb_key`: same for $\omega = 4$ type $(1, 1, 2)$ (zero sorry).
- `pasten_F31a`: $(a + 2)^a < (2a)^{a+4}$ for all $a \geq 1$; algebraic core of the Quality- ρ Joint Bound (zero sorry).
- `pasten_quality_lt_half_core`: $a + b < ab$ for $a \geq 2, b \geq 3$; integer core of quality $< 1/2$ (zero sorry).
- `pasten_F32a_gap`: $q + 2 \leq k \cdot q$ for all $k \geq 2, q \geq 3$; no non-degenerate norm lies strictly in $(q, q + 2)$ (F32a core, zero sorry).
- `pasten_F32a_upper`: $\max(\max(0, q), q + 2) = q + 2$; norm of the second-minimum vector $\varphi = (0, 1, 1)$ (zero sorry).
- `pasten_F32b_ineq`: $2q + 4 \leq q^2$ for $q \geq 5$; algebraic core blocking p -term dominance in the F32b spectrum proof (zero sorry).
- `pasten_F33_gap`: $N \leq k \cdot r_1$ for $k \geq 2, N \leq 2r_1$; integer core of second-minimum gap for type $(1, 1, 2)$ (F33, Theorem 33, zero sorry).
- `pasten_En_lb_general`: for any Finsets P_a, P_b, P_c of primes satisfying the lattice constraint $\sum_{P_a} f + \sum_{P_b} f = \sum_{P_c} f$, if $f = 0$ on P_b and $f = 0$ on P_c , then $\sum_{P_b} f = \sum_{P_a} f$ (degenerate condition for all ω ; Lean, zero sorry). This is the first fully general- ω Lean formalization in this project.
- Four OB-13B integer kernel lemmas (zero sorry): cross-group norm bound; $\text{median}^{\omega-1} \leq R$ for $\omega = 3, 4$; within-group $W \neq 0$; all in `lean/AbcHeightKernel.lean`.
- `pasten_nd_lb_large_prime_zero`: $p_2 \leq p, px < p_2 \Rightarrow x = 0$; integer core of LB-1 (zero sorry).
- `pasten_nd_lb_single_prime_int`: $k > 0, kz = 0 \Rightarrow z = 0$; abstract form of Lemma 18 (zero sorry).
- `pasten_nd_lb_omega2_ge_q`: $q \leq q(v/g)$ for $g \mid v$; integer core of $\text{nd} \geq p_2$ in Theorem 20 (zero sorry).
- Three type $(k, 1, 1)$ integer kernel lemmas for Theorem 22 (zero sorry): `pasten_nd_k11_r_branch` ($r \leq rx$ for $x \geq 1; \varphi_r \neq 0$ branch); `pasten_nd_k11_valuation_branch` ($qk \leq qkx; \varphi_r = 0$ branch); `pasten_nd_k11_norm_equals_qk` ($\max(px, qkx) = qkx$ for $p < q$; q -component dominates).

- `pasten.nd.km1.gcd.div.one`: $\gcd(k/g, m/g) = 1$ for $g = \gcd(k, m)$; integer core of the $\varphi_r=0$ solution-family uniqueness in Theorem 23 (Mathlib `Nat.Coprime.div.gcd.coprime`, zero sorry).
- `pasten.nd.kmn.val.witness.constraint`: $k \cdot (m/g) = m \cdot (k/g)$ for $g = \gcd(k, m)$; confirms the valuation-regime witness $(-m/g, k/g, 0)$ satisfies the constraint for *any* n (Corollary 24), using `Nat.mul.div.assoc` twice. Zero sorry.
- `pasten.nd.kmn.val.n.zero`: $n \cdot 0 = 0$ for all n ; captures the key n -independence: the r -coordinate of the valuation-regime witness is zero, so the c -exponent n does not appear (Corollary 24). Zero sorry.

7 Conclusion

We have proved that for squarefree coprime triples (a, b, c) with $a + b = c$ and $\omega(abc) \geq 2$, the Pasten lattice satisfies $\det(F(a, b)) < \text{rad}(abc)$, and hence Minkowski's theorem [1] gives a nonzero vector with $\|\psi\|_\infty \leq \text{rad}(abc)^{1/(\omega-1)}$.

Beyond the determinant bound, we have established a complete classification of non-degeneracy for all partition types with $\omega \leq 5$, proved that all types with $\omega \geq 6$ are unbounded (Theorem 34), and determined the exact sharp bounds for every universally-bounded type (Theorems 35 and 36). The central structural results are: (a) the Universal Divisibility Lemma ($q \mid \psi_q$ for all $q \in P$, proved by elementary divisibility from the lattice constraint); (b) the φ -coordinate change reducing $F(a, b)$ to the simplest integer hyperplane lattice; (c) the optimal crossing formula (Theorem F10): the minimum non-degenerate norm is the second smallest of $\{\min P_a, \min P_b, \min P_c\}$, achieved by any cross-group prime pair; (d) the universal lower bound (Theorem E_n): every non-degenerate ψ satisfies $\|\psi\|_\infty \geq \text{nd}$, proved by a clean F8+constraint+Wronskian argument and Lean-formalized for $\omega = 3, 4$ (E10/E11), numerically verified for all types $\omega \leq 7$; (e) a complete table of sharp non-degenerate minimum norms for all partition types $\omega \in \{3, 4, 5\}$ (Table in Theorem 35); (f) the unifying pattern $\sup_{\text{balanced}}(\omega) = 2^{-1/(\omega-1)}$ for *all* $\omega \geq 3$ (Theorem 36, F28), with the general key lemma Lean-formalized; and (g) the general bound $\rho^4 < 1/6$ for *all* type $(2, 2, 1)$ triples (F29, Lean-formalized), with $\sup = 6^{-1/4}$ approached by the $a = 6$ near-twin-prime subfamily; and (h) the corrected bound $\rho^4 < 1/2$ for *all* type $(2, 1, 2)$ triples (F30, Lean-formalized, correcting F25), with $\sup = 2^{-1/4}$ equal to the $(1, 2, 2)$ value by $a \leftrightarrow b$ symmetry; and (i) the Quality- ρ Joint Bound (F31, Theorem 5): for every non-trivial squarefree triple of any bounded- ρ type, $\text{quality} + \rho^{\omega-1} < 1$, proved by combining $\text{quality} < 1/2$ (Proposition 4) with $\rho^{\omega-1} < 1/2$ (Theorems F12–F30); and (j) the second-minimum and complete spectrum results for type $(1, 1, 1)$: the second minimum non-degenerate norm equals r (F32a, Theorem 30), and for every twin-prime triple with $q \geq 5$ the full spectrum of achievable non-degenerate norms is exactly $\{kq : k \geq 1\} \cup \{kr : k \geq 1\}$ (F32b, Theorem 31), both Lean-formalized; the Frobenius number of the spectrum is $q^2 - 2$ (Corollary 32); and second minimum $\min(r_2, q, 2r_1)$ for type $(1, 1, 2)$ (F33, Theorem 33), Lean-formalized; and (k) the non-squarefree upper bound (OB-13B, Theorem 15): $\text{nd}(a, b) \leq v_{\max} \cdot R^{1/(\omega^*-1)}$ for all coprime triples, proved via the cross-group vector construction, with a tighter median form (Corollary 16), and the within-group construction

showing that the squarefree E_n lower bound fails for non-squarefree triples (counterexample (72, 11, 83)); (l) the universal lower bound (Theorem 19): $\text{nd}(a, b) \geq p_2$ (second-smallest prime in $\text{rad}(abc)$) for all coprime triples, squarefree or not, proved by a single-prime-constraint argument (Lemma 18), tight at (1, 2, 3); and (m) the exact formula for $\omega^* = 2$ triples (Theorem 20, Corollary 21): $\text{nd}(a, b) = \max(p \cdot w/g, q \cdot v/g)$ where $g = \gcd(v, w)$, verified on all seven $\omega^* = 2$ examples up to $c \leq 1000$, Lean-formalized; and (n) the exact formula for type $(k, 1, 1)$ non-squarefree triples (Theorem 22): $\text{nd}(a, b) = \min(r, qk)$ for $P_a = \{p^k\}$, $P_b = \{q\}$, $P_c = \{r\}$ with $p < q < r$, with two regimes (valuation-dominant $qk < r$ and pairwise-dominant $qk \geq r$) and confirming $\text{nd} \geq p_2 = q$ (Theorem 19); verified numerically (T68) and Lean-formalized (three arithmetic kernel lemmas, zero `sorry`); and (o) the exact formula for general type $(k, m, 1)$ non-squarefree triples (Theorem 23): $\text{nd}(a, b) = \min(r, \max(pm/g, qk/g))$ where $g = \gcd(k, m)$, subsumes Theorems 22 ($m = 1$) and the squarefree E_n value ($k = m = 1$), with equal-valuation regime ($k = m$) always realising $\text{nd} = p_2$; verified numerically on 14 named examples (T69); and (p) a universality corollary (Corollary 26): for one-prime-per- a, b -group triples with $r_1 = \min P_c > q > p$, the formula $\text{nd} = \min(r_1, \max(pm/g, qk/g))$ holds regardless of how many primes c has or their valuations — only the smallest c -prime matters; and (q) a valuation-regime independence result (Corollary 24): when $\max(pm/g, qk/g) \leq r$, the formula $\text{nd}(a, b) = \max(pm/g, qk/g)$ holds for *all* $n = v_r(c)$, not just $n = 1$ — the witness $(-m/g, k/g, 0)$ satisfies the constraint for any n , so the c -exponent is entirely invisible in the valuation regime (verified numerically on 52 cases with $n \geq 2$, T71); and (r) a characterisation of the pure-formula boundary (Remark 8): when the group- c prime is not the largest in the triple, the three-pure formula can fail and a fully mixed witness is required, exemplified by (5, 27, 32) where $\text{nd} = 6$ beats all pure-branch minima ($N_0 = N_1 = 15$, $N_2 = 25$); and (s) a complete pairwise-regime characterisation (Theorem 25): for one-prime-per-group triples in the pairwise regime $N_{p_L=0} > p_L$, the exact formula is $\text{nd}(a, b) = \min(N, \max(p_L, B))$ where B is the minimum weighted norm over $|\varphi_{p_L}| = 1$ solutions — simultaneously giving $\text{nd} = p_L$ in the typical case ($B \leq p_L$, verified 223/229 cases, T72) and recovering the precise value in the 6 exceptional cases (e.g. (169, 343) with $\text{nd} = N = 21$ due to a gcd obstruction); and (t) a systematic scan (Remark 10, T73) of all quality- > 1 one-prime-per-group triples with $a + b \leq 5 \times 10^5$, yielding exactly 10 such triples — all correctly classified by the exact formula, with four exhibiting $\text{nd} < p_L$ via the $\text{val}(\varphi_{p_L} = 0)$ regime, and confirming that nd and quality are structurally independent (nd depends on (p_a, p_b, p_c, k, m) while quality depends on n); and (u) the exact nd formula for $\omega^* = 4$ triples with two c -group primes (Theorem 27): for $P_a = \{p\}$, $P_b = \{q\}$, $P_c = \{r, s\}$, $\text{nd}(a, b) = \min(\text{nd}_3(p^k, q^m, r^j), \text{nd}_3(p^k, q^m, s^n), \max(q, B_m), \max(p, B_k))$ where B_x is the two-dimensional Bezout minimum for $j\varphi_r + n\varphi_s = x$; the $\varphi_q = 0$ branch wins in 25 out of 431 tested cases (T74), confirming that the four-branch formula subsumes the $\omega^* = 3$ characterization; and (v) the exact nd formula for type (2, 1, 1) triples (Theorem 28): for $P_a = \{p, q\}$, $P_b = \{r\}$, $P_c = \{s\}$, the minimum is taken over ten branches covering all zero-coordinate patterns — within-group W , five pairwise Bezout norms, two P_a -interaction norms $N_{r=0}$ and $N_{s=0}$, and two $\omega^* = 3$ sub-problems; verified on 672 triples (T77, 0 failures), with $N_{s=0}$ dominant in 87% of cases; and (w) the exact nd formula for type (1, 2, 1) triples (Theorem 29): for $P_a = \{p\}$, $P_b = \{q, r\}$, $P_c = \{s\}$, the same ten-branch structure applies with $P_a \leftrightarrow P_b$ exchanged — within-group W_b (valid iff $j \neq m$), five pairwise norms, two P_b -interaction norms $N_{p=0}$ and $N_{s=0}$, and two $\omega^* = 3$ sub-problems; verified on 1 043 triples

with $a, b \leq 200$ (T78, 0 failures), with cross-group dominant in 61% and W_b in 30%.

The non-degenerate ratio $\|\psi_{\text{nd}}\|/R^{1/(\omega-1)}$ is bounded for exactly the “balanced” partition types (both other constituents non-empty with available small-prime crossings), and grows without bound for the “isolated” types where both non-dominant constituents are forced-large single primes. For $\omega \geq 6$, every type is unbounded. The sharp bounds for the bounded types are now analytically complete (Theorem 35), with exact suprema proved for $\omega \in \{3, 4, 5\}$, the universal formula $2^{-1/(\omega-1)}$ proved for all balanced single-prime-group types (Theorem 36), the general bound $\rho^4 < 6^{-1}$ proved for all type $(2, 2, 1)$ triples with $\text{sup} = 6^{-1/4}$ (F29), and the general bound $\rho^4 < 1/2$ proved for all type $(2, 1, 2)$ triples with $\text{sup} = 2^{-1/4}$ (F30). Both bounds are Lean-formalized.

We also prove a quality boundary for the squarefree subfamily (Proposition 4): quality $> 1/2$ if and only if $a = 1$. Within bounded partition types, quality and ρ are negatively correlated ($r \approx -0.76$ for types $(1, 1, 2)$ and $(1, 2, 1)$): the triples closest to the sharp ρ bound have quality $\rightarrow 1/3$, while the highest-quality non-trivial squarefree triples ($a = 2, b \rightarrow \infty$) have $\rho \rightarrow 0$. This quality- ρ trade-off is a provable structural feature, not a numerical artifact.

Honest scope. None of these results constitute progress toward the abc conjecture or SDC. The Minkowski bound is $\eta_R = 1/(\omega - 1)$ relative to $\text{rad}(abc)$, whereas SDC requires $\eta_c < 1$ relative to c for *all* $\varepsilon > 0$ — an exponential improvement. The classification of degenerate vs. non-degenerate types is structural geometry of Pasten’s lattice; it does not produce bounds on c . The abc conjecture remains open.

This is a structural result about the geometry of Pasten’s lattice construction, not progress toward abc. The abc conjecture requires $\|\psi\|_\infty = o(c^\varepsilon)$ for all $\varepsilon > 0$ (Pasten’s SDC), and the gap between our bound and SDC is the full depth of the problem. We have made no use of any abc-equivalent hypothesis, known abc triples, or the Szpiro conjecture.

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